

Laís A. Grossel

Simpósio: “Next Generation Opilionologist”,
investigaciones de punta en opilionología
desarrolladas por jóvenes talentos



Contribuciones de un opilión Neotropical a
nuestra comprensión de la interacción entre
la **selección sexual** y el **cuidado paterno**

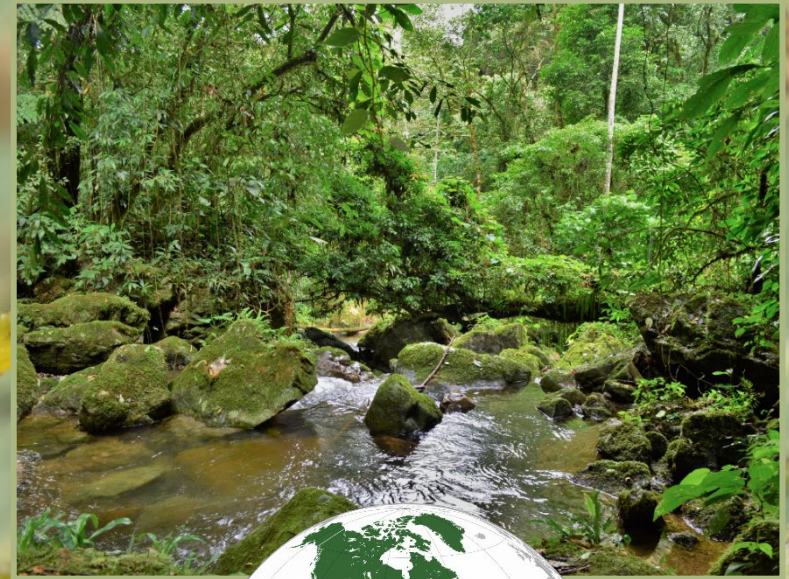
Iporangaia pustulosa
Mello-Leitão, 1935



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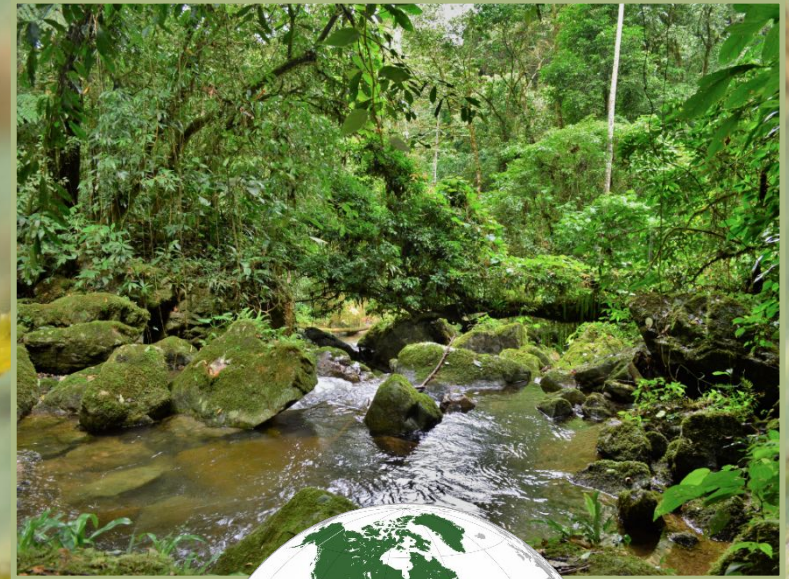


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Sociobiology Vol. 44, No. 3, 2004

**Five New Cases of Paternal Care in Harvestmen
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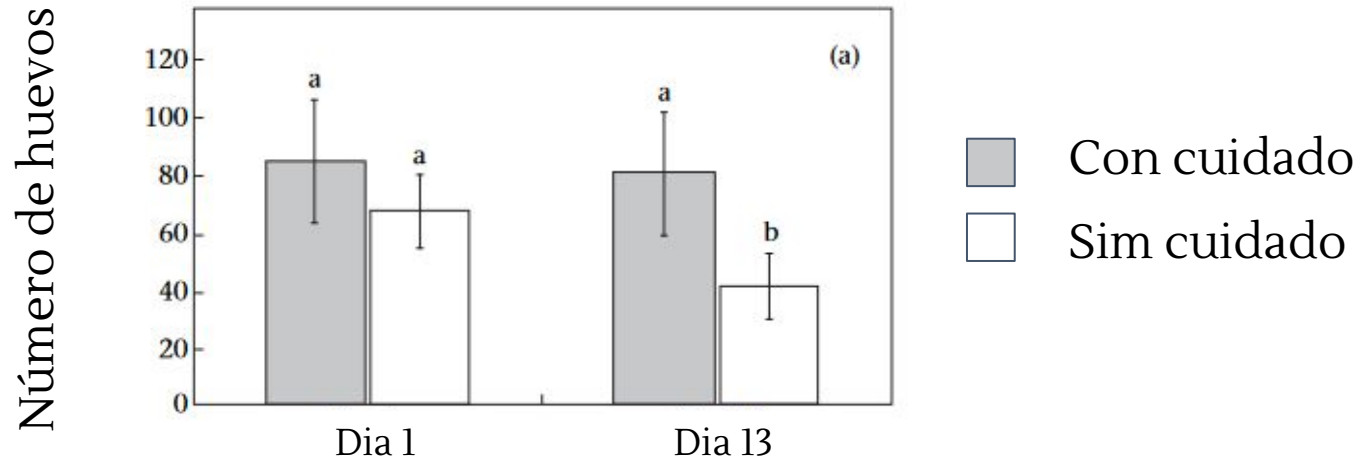


Efficiency of uniparental male and female care against egg predators in two closely related syntopic harvestmen

Gustavo S. Requena^a, Bruno A. Buzatto^b, Roberto Munguía-Steyer^c, Glauco Machado^{c,*}

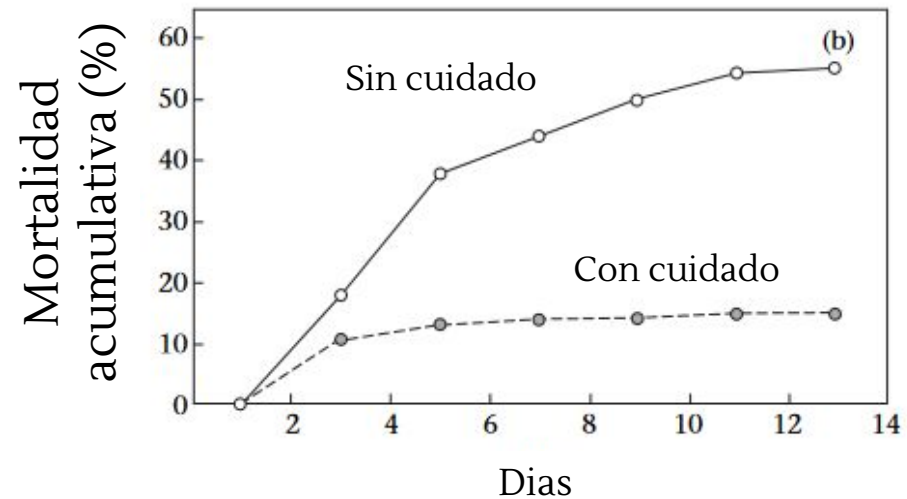
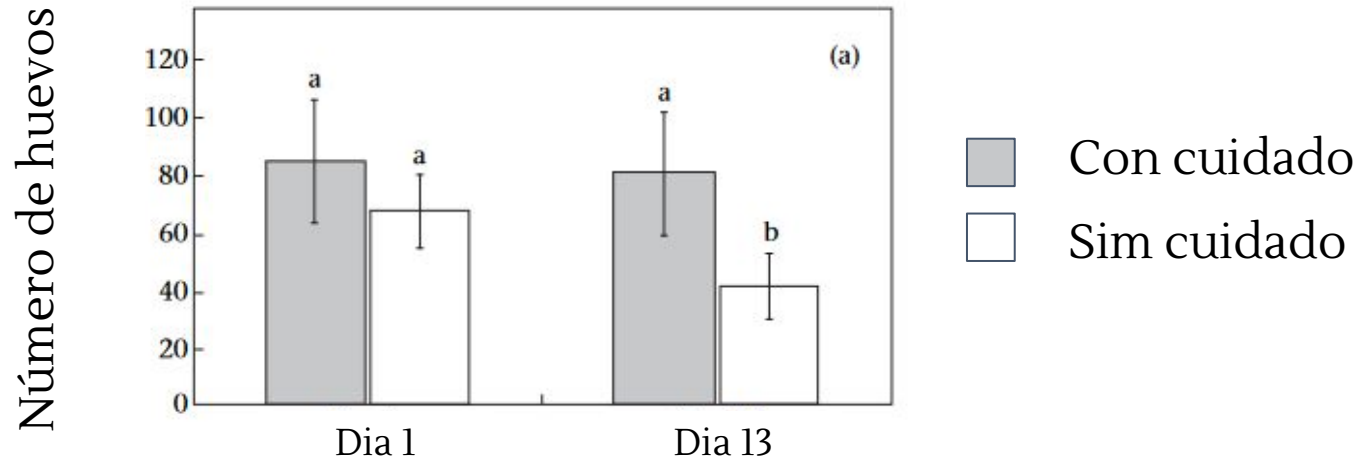
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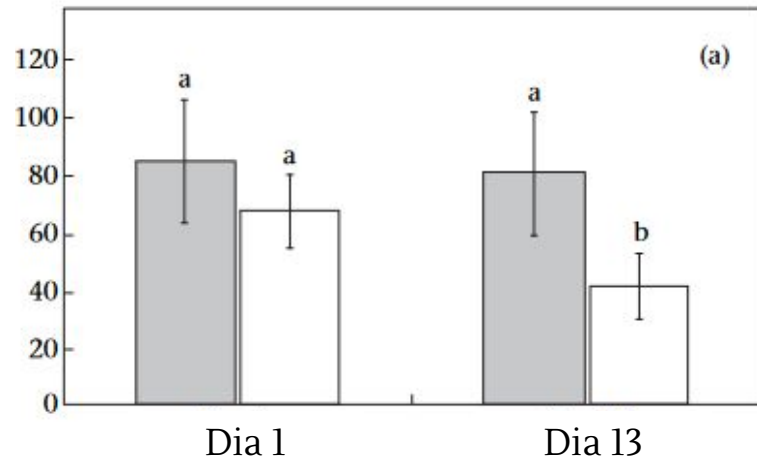
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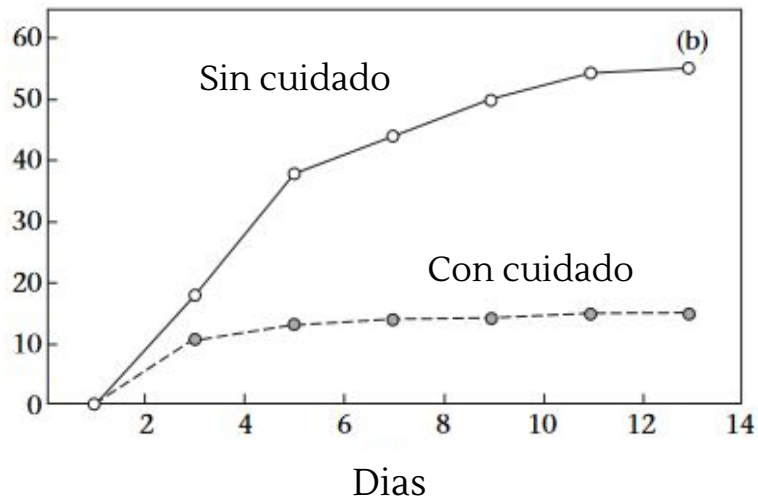
Número de huevos



■ Con cuidado
□ Sin cuidado

- Una **reducción** promedio del 55% en el número de huevos;

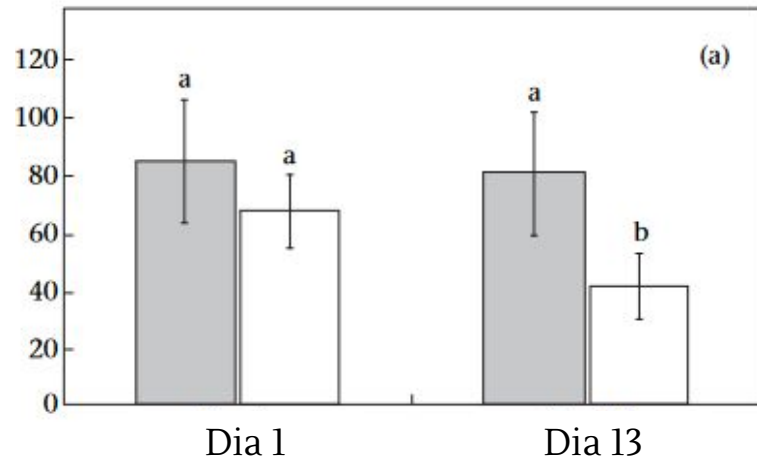
Mortalidad acumulativa (%)



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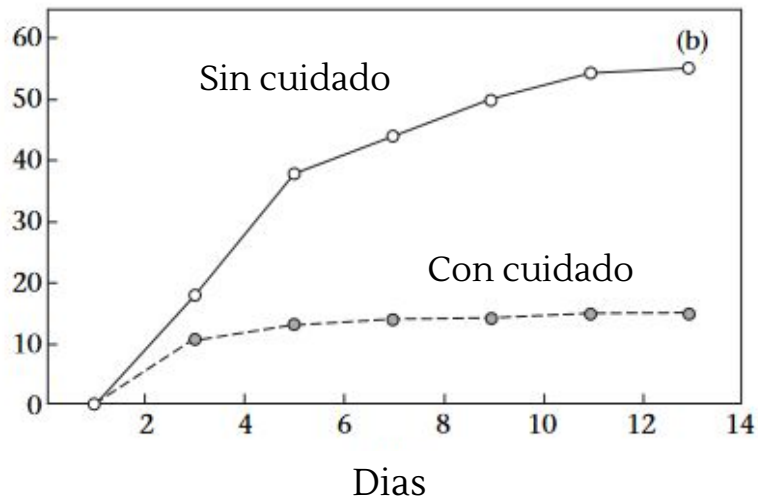
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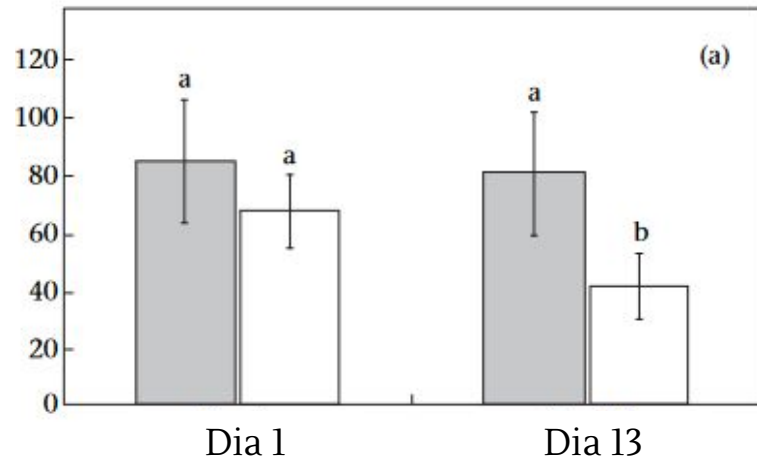


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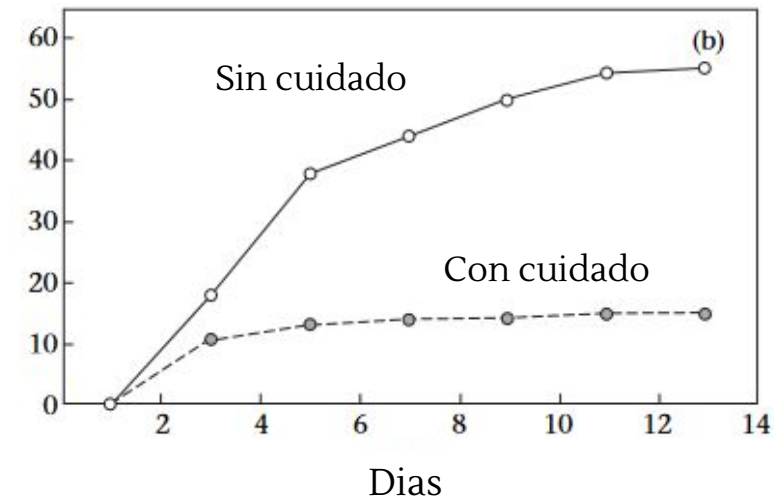
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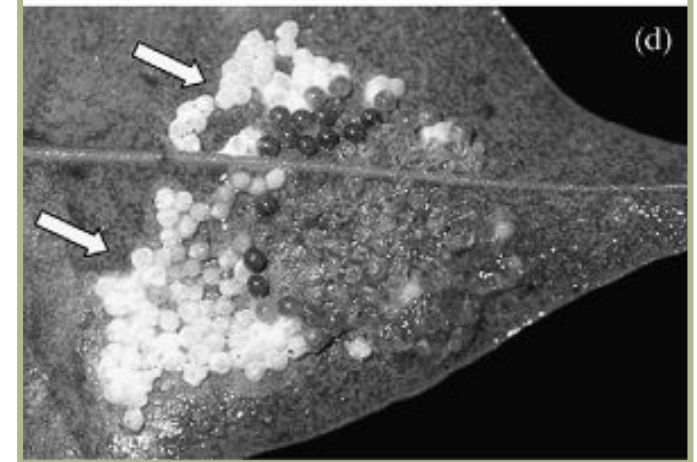
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Ajuste del cuidado paterno

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Beneficio del
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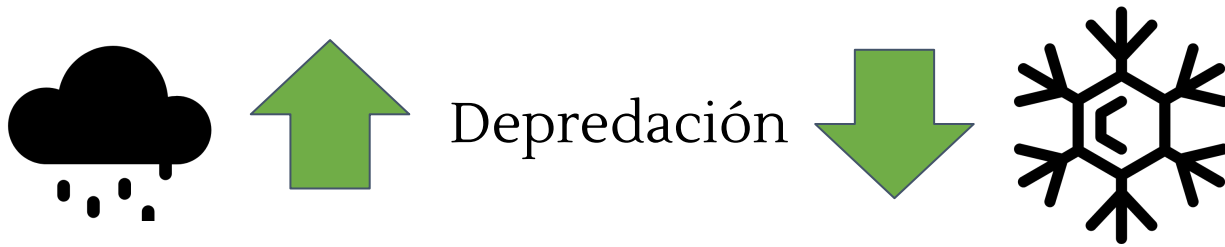
Beneficio del
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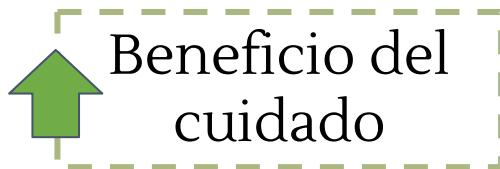
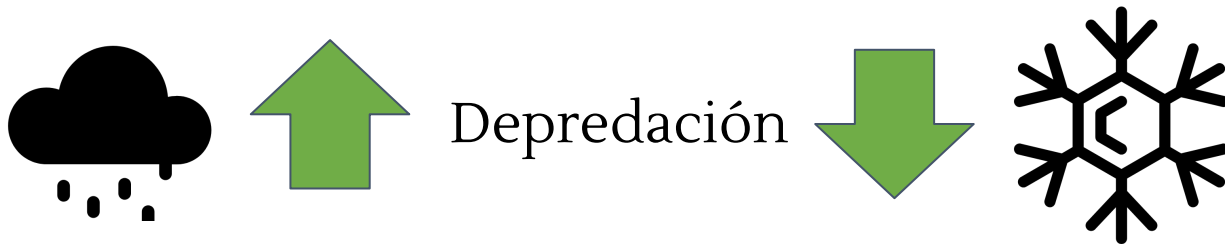
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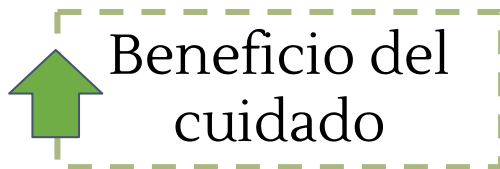
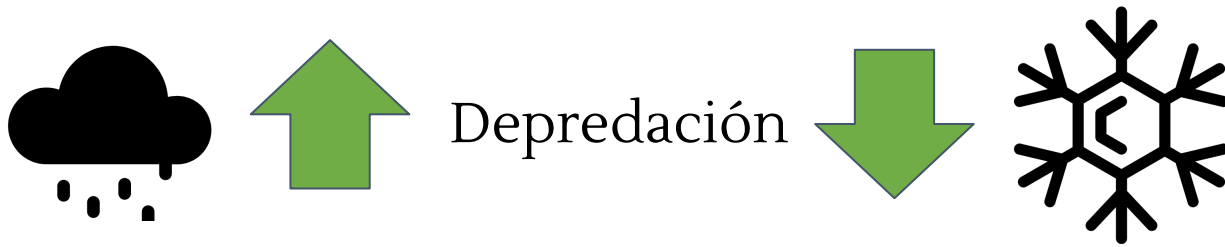


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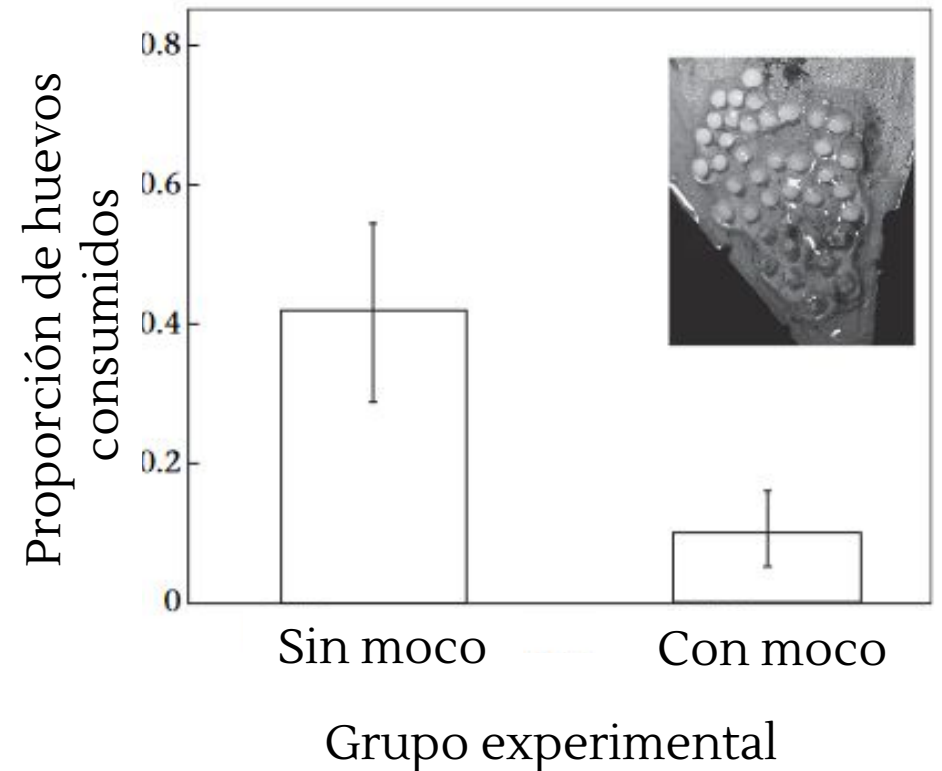
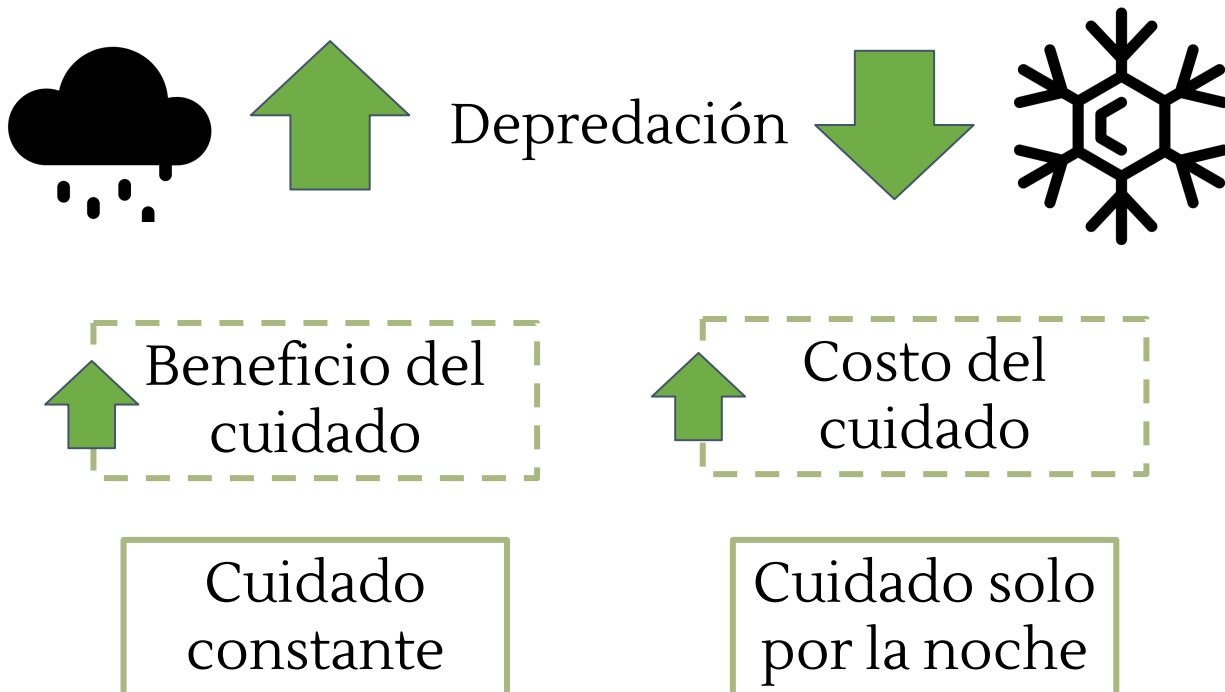


Cuidado solo
por la noche

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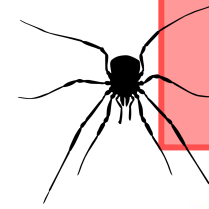
Ajuste del cuidado paterno



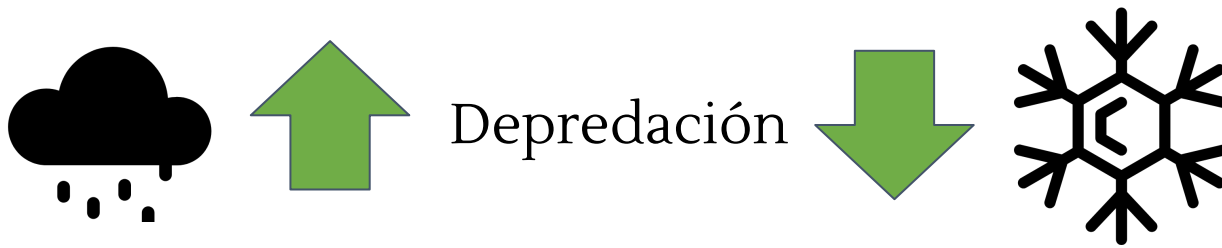
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Cuidado paterno aumenta la supervivencia de la puesta



Ajuste del cuidado paterno

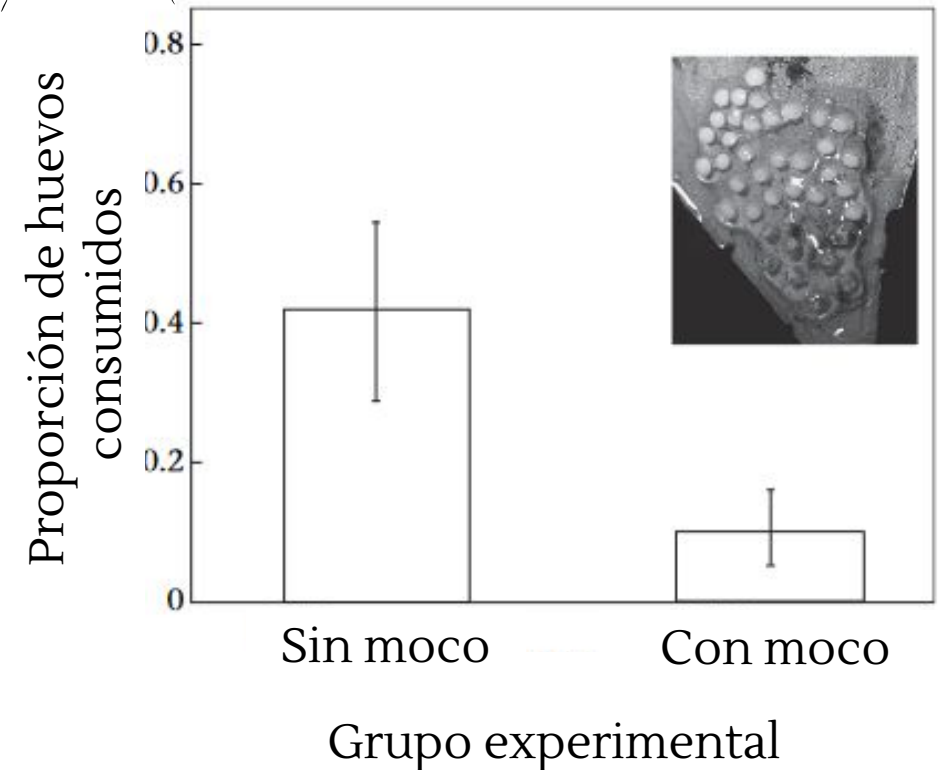


↑
Beneficio del
cuidado

Cuidado
constante

↑
Costo del
cuidado

Cuidado solo
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Paternal Care Decreases Foraging Activity and Body Condition, but Does Not Impose Survival Costs to Caring Males in a Neotropical Arachnid

October 2012 | Volume 7 | Issue 10 | e46701

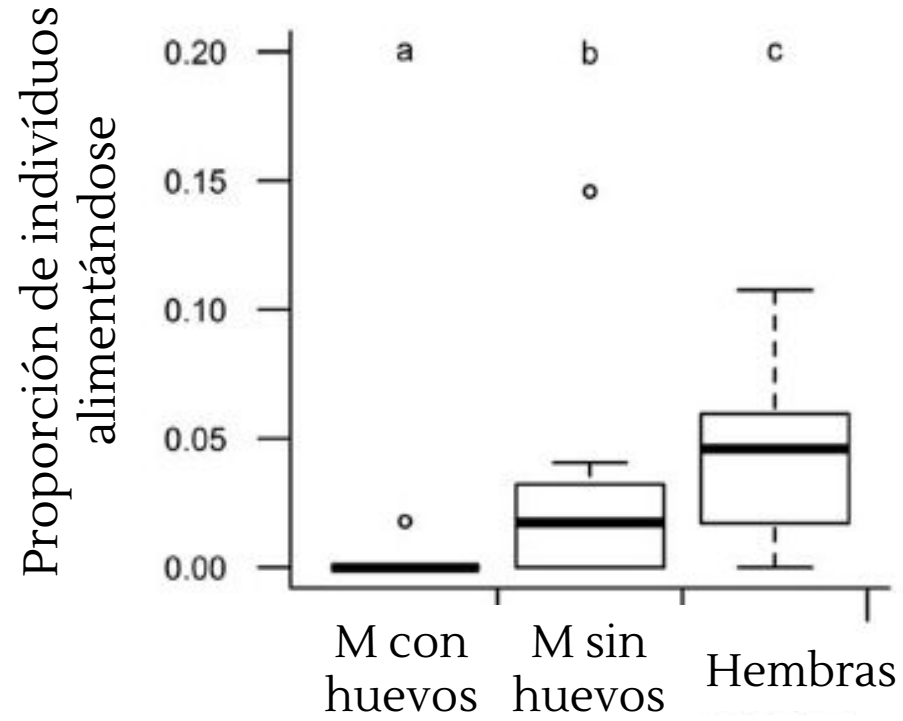
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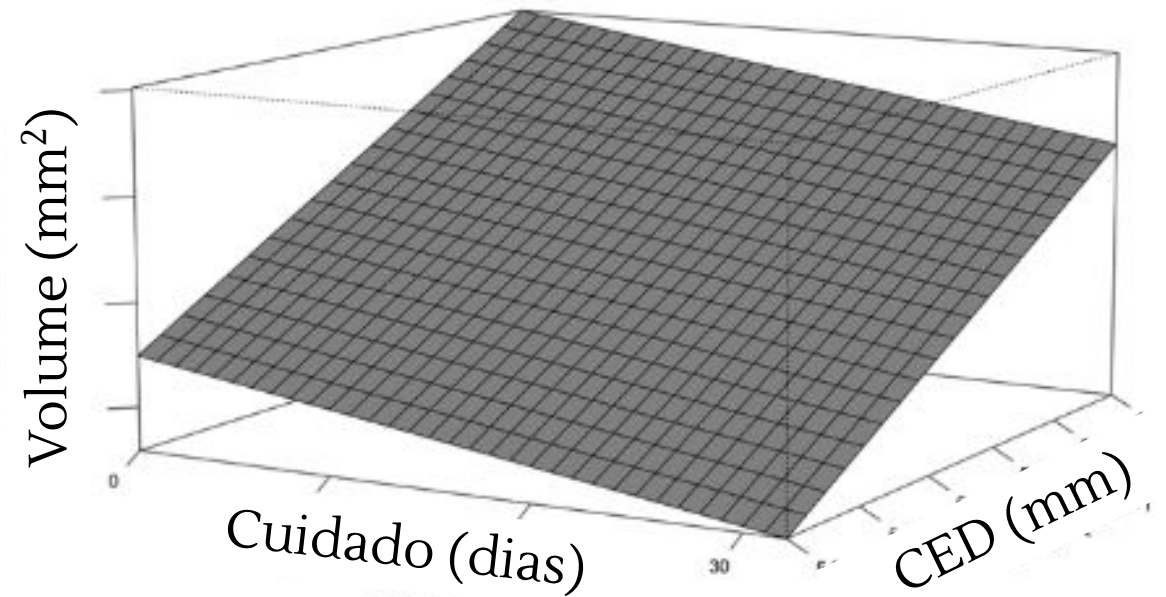
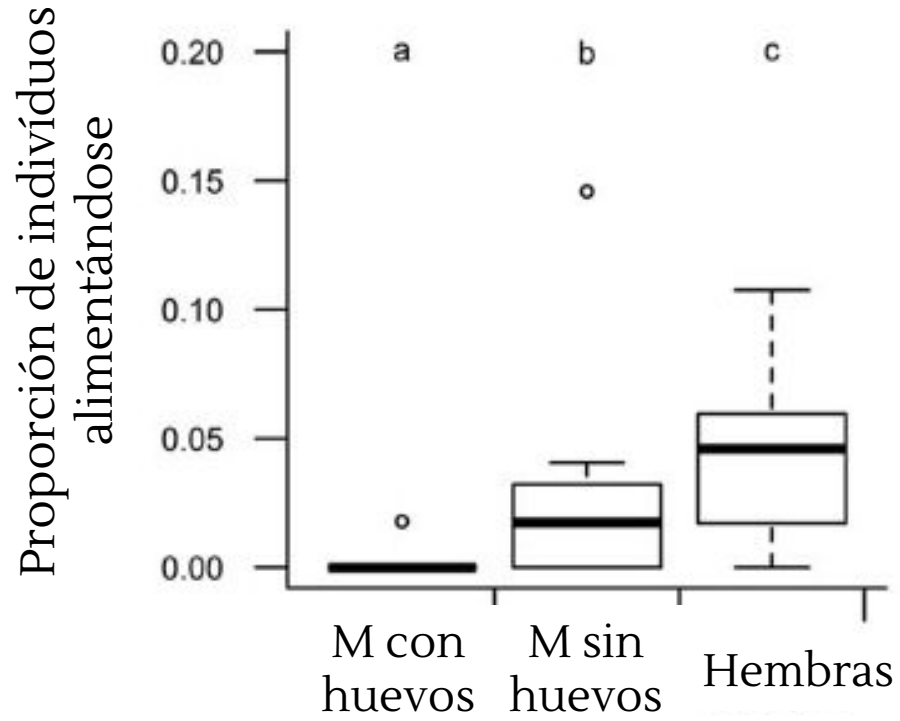
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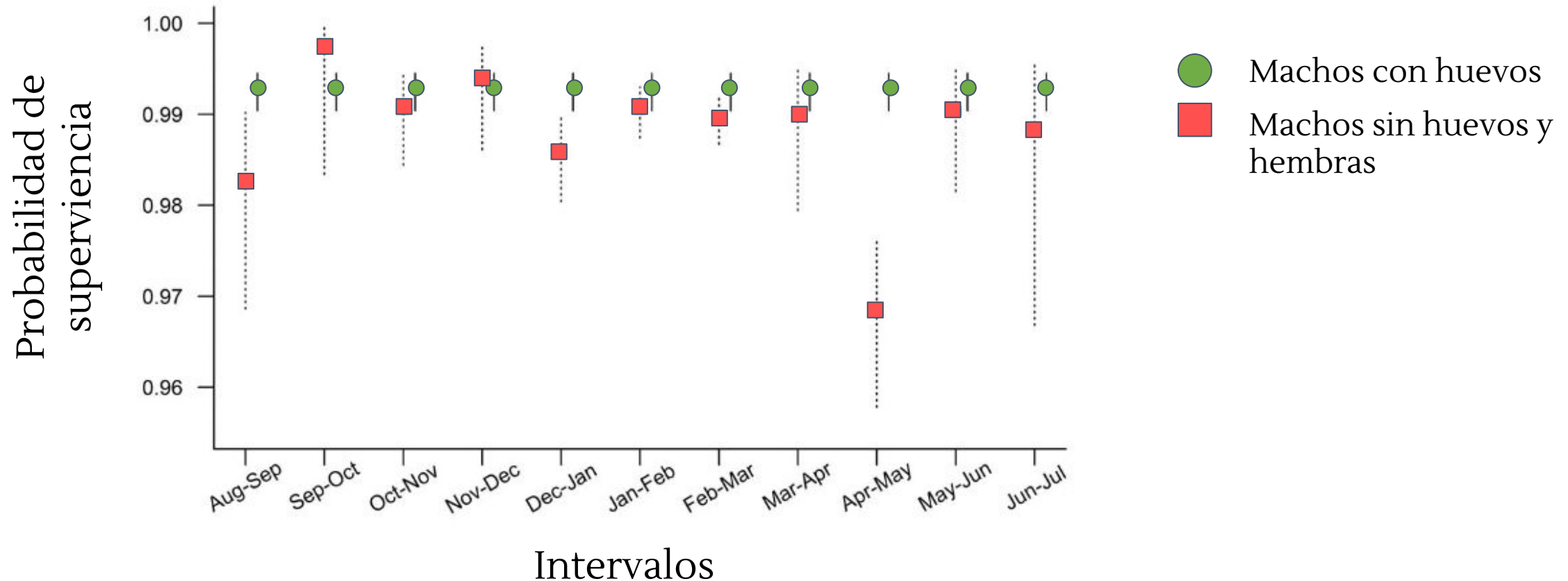
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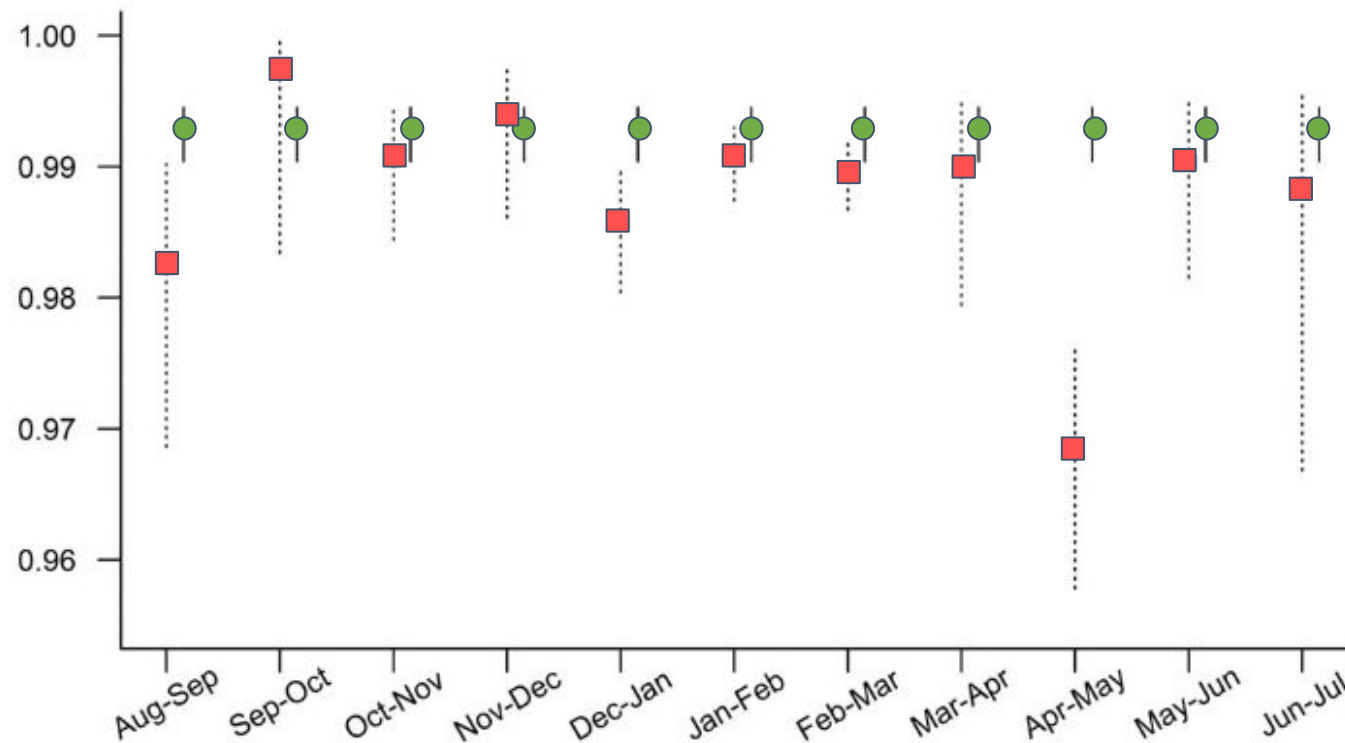
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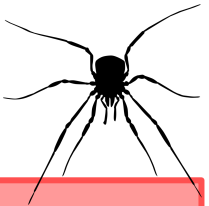


Probabilidad de supervivencia



Intervalos

● Machos con huevos
■ Machos sin huevos y hembras



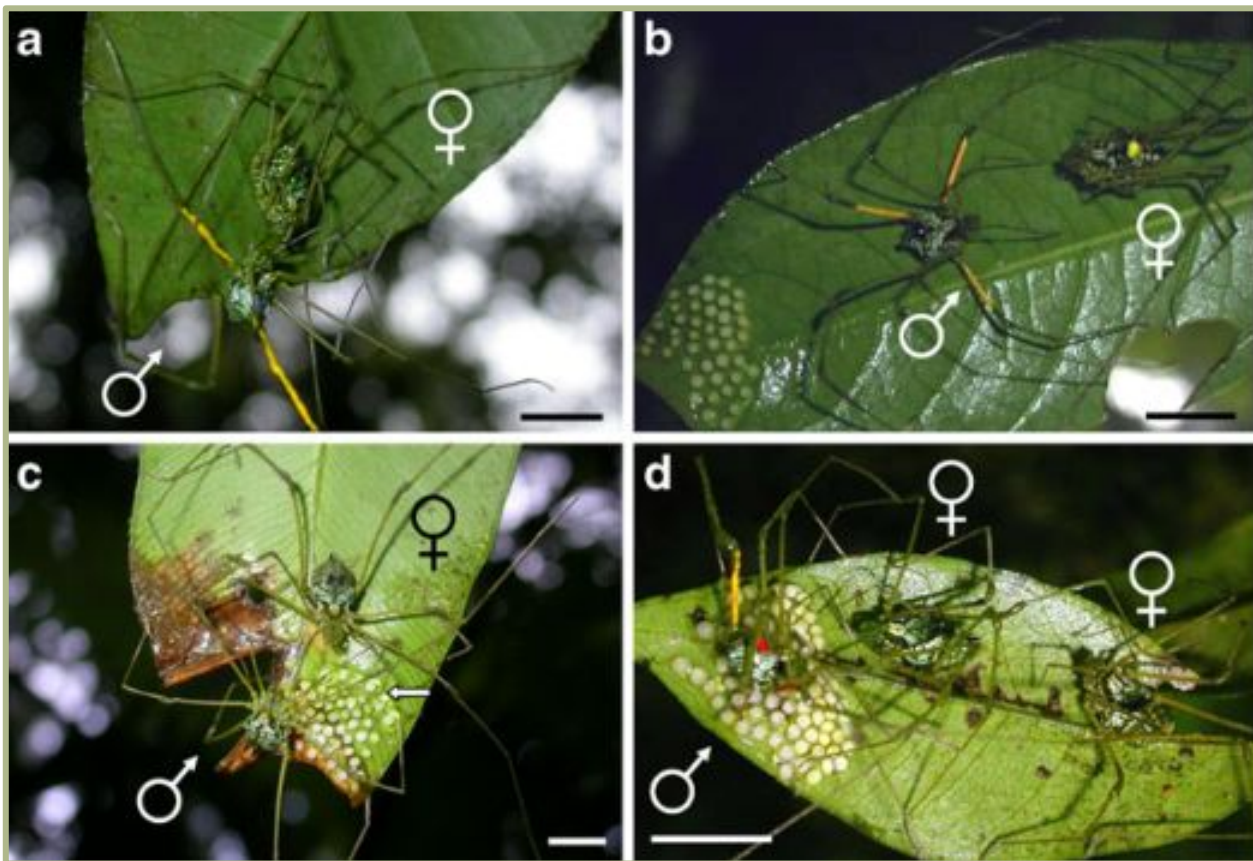
Reducción de los riesgos de mortalidad favorece a los machos parentales

Mating behavior of a Neotropical arachnid with exclusive paternal care

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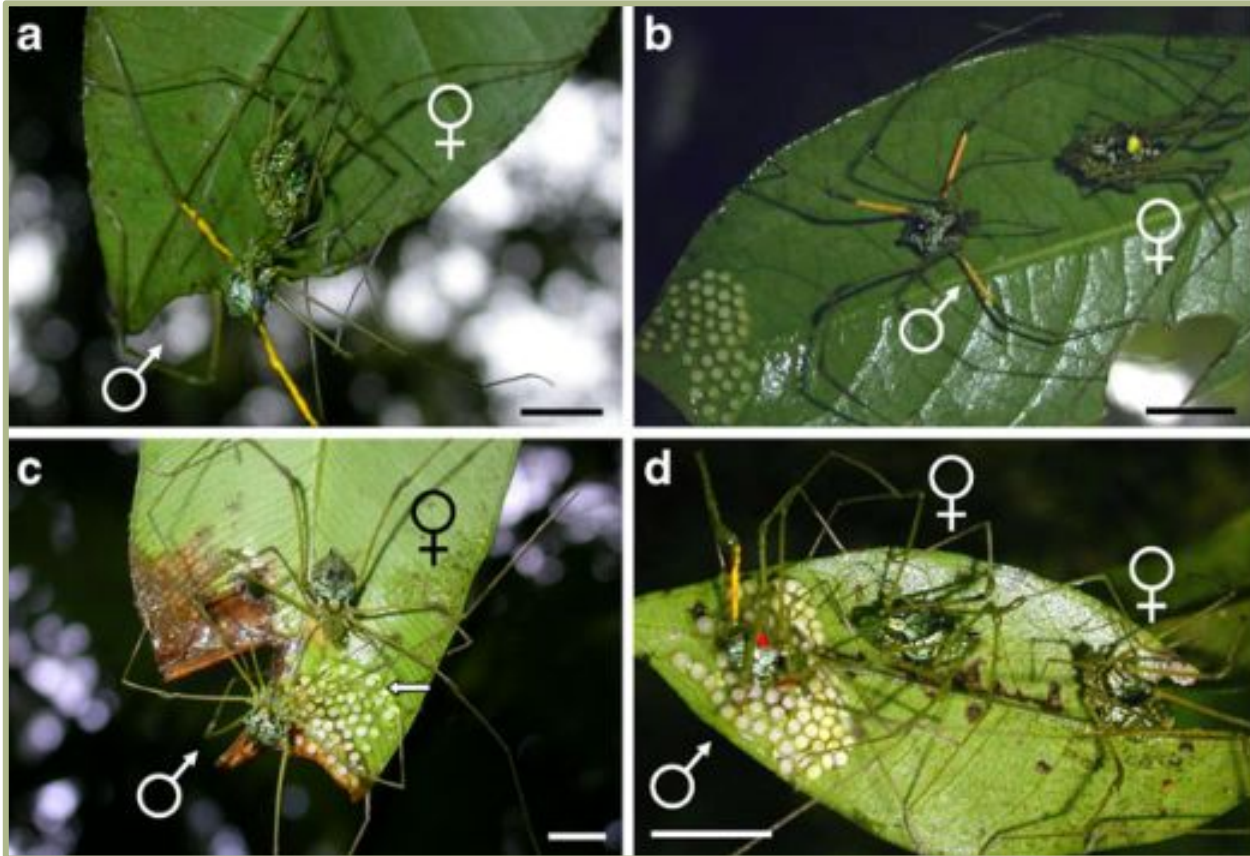
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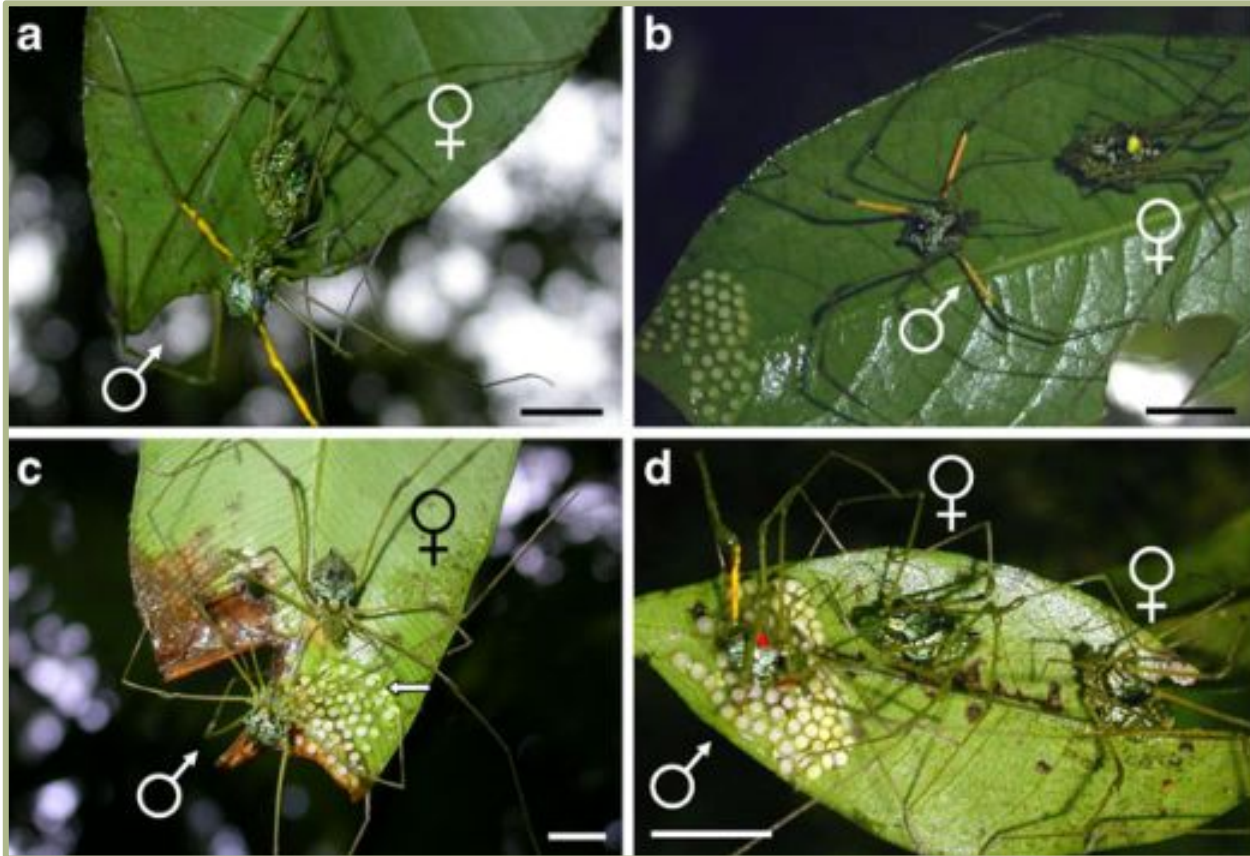
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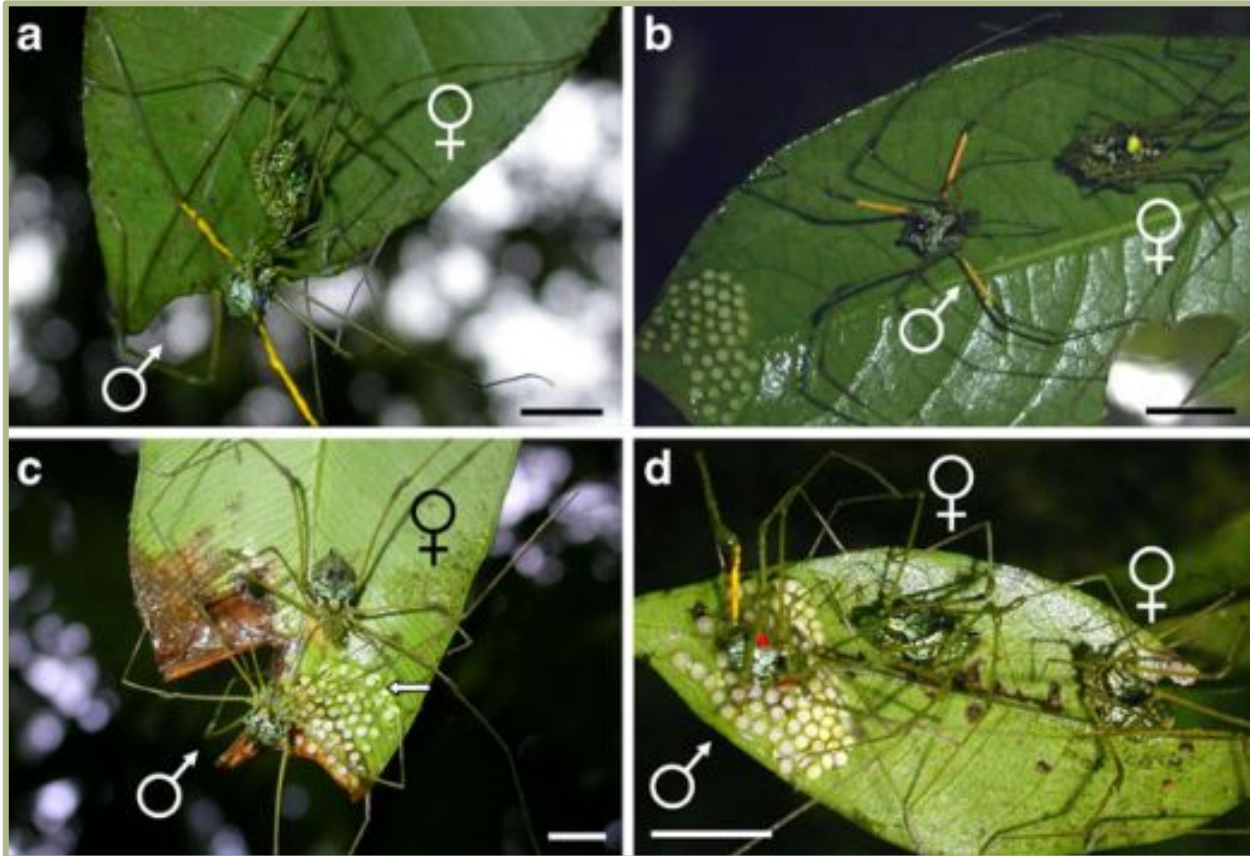
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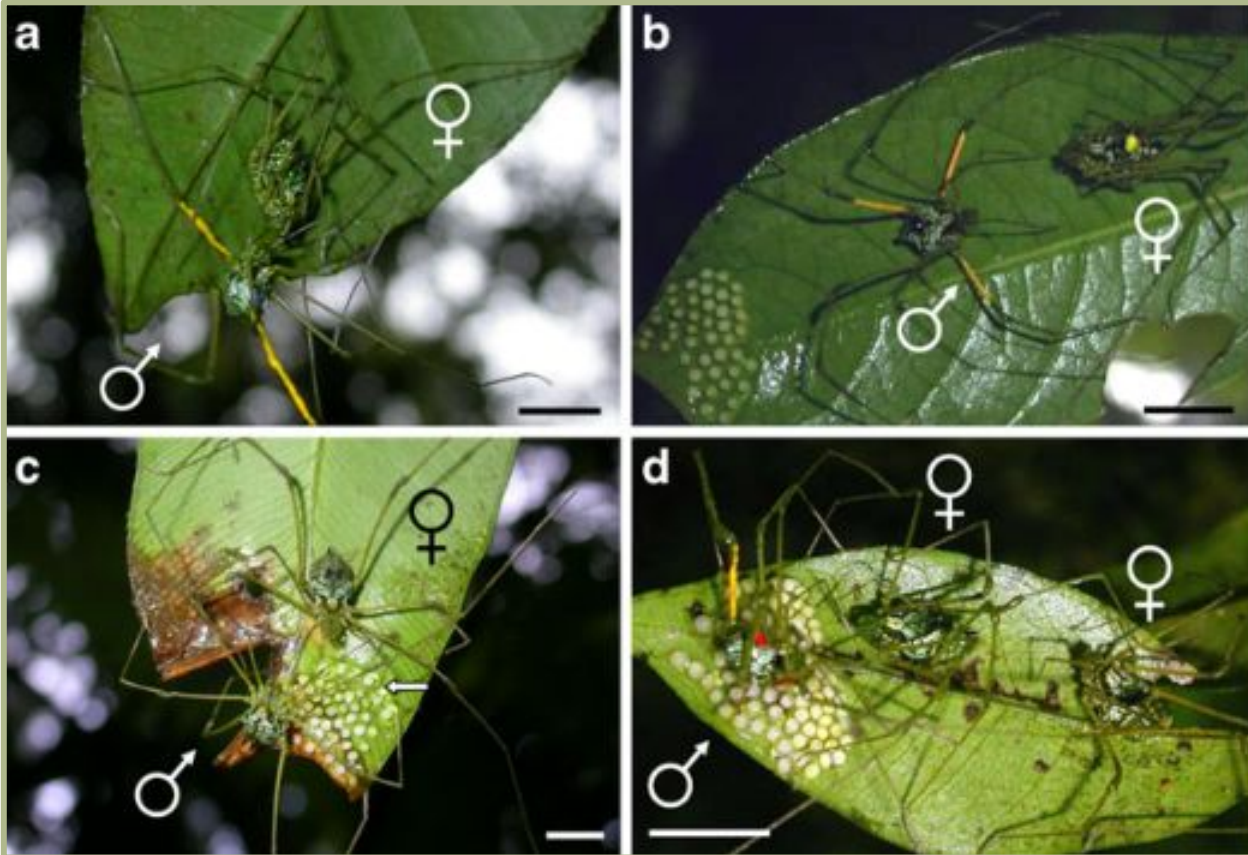
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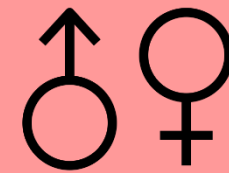
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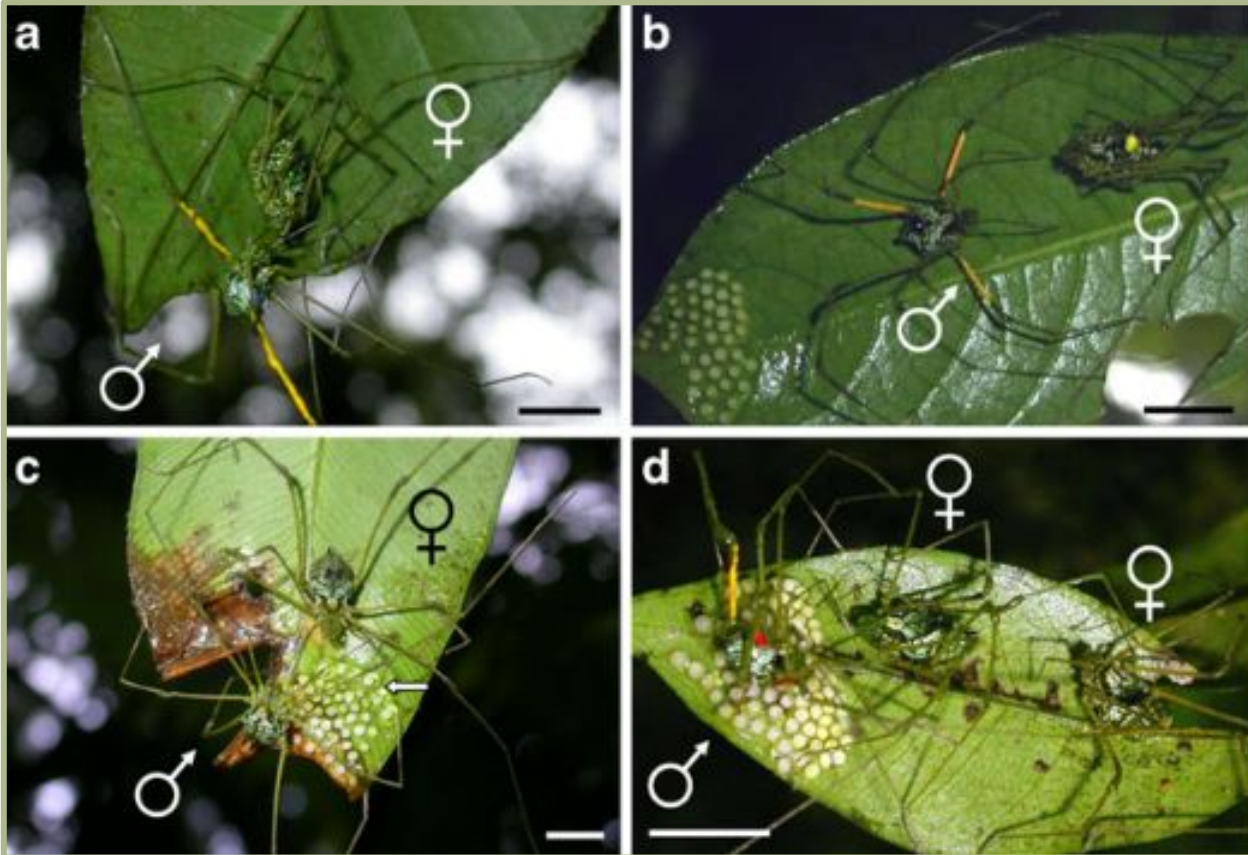


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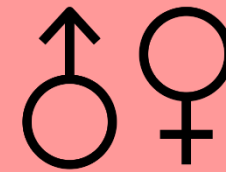
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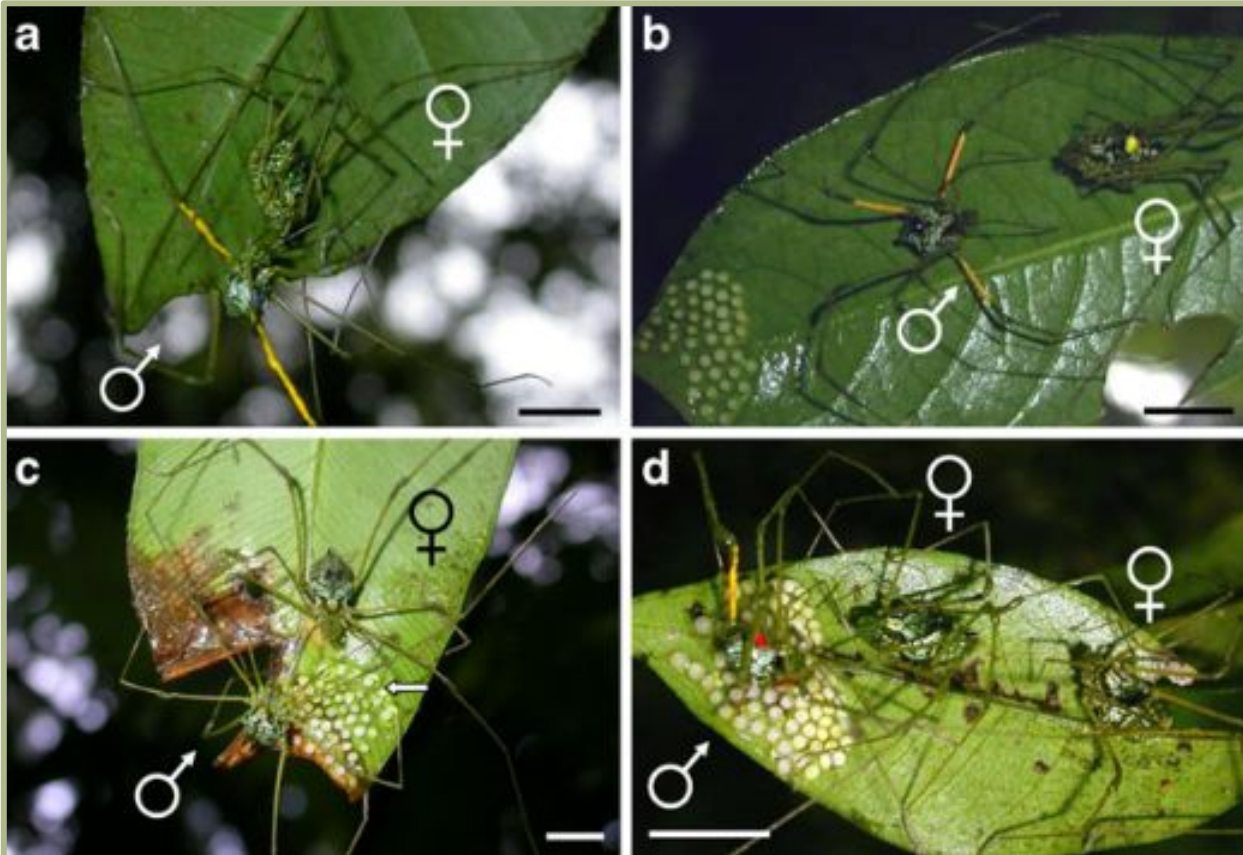
Rechazo
agresivo



Abandono
sin
oviposición

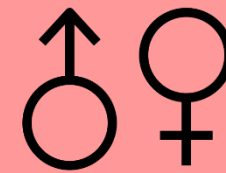
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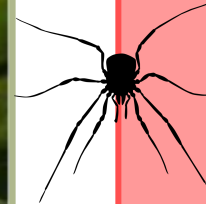
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Rechazo
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Abandono
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**Ambos sexos pueden
evaluar y seleccionar parejas**



Effects of egg attendance on male mating success in a harvestman with exclusive paternal care

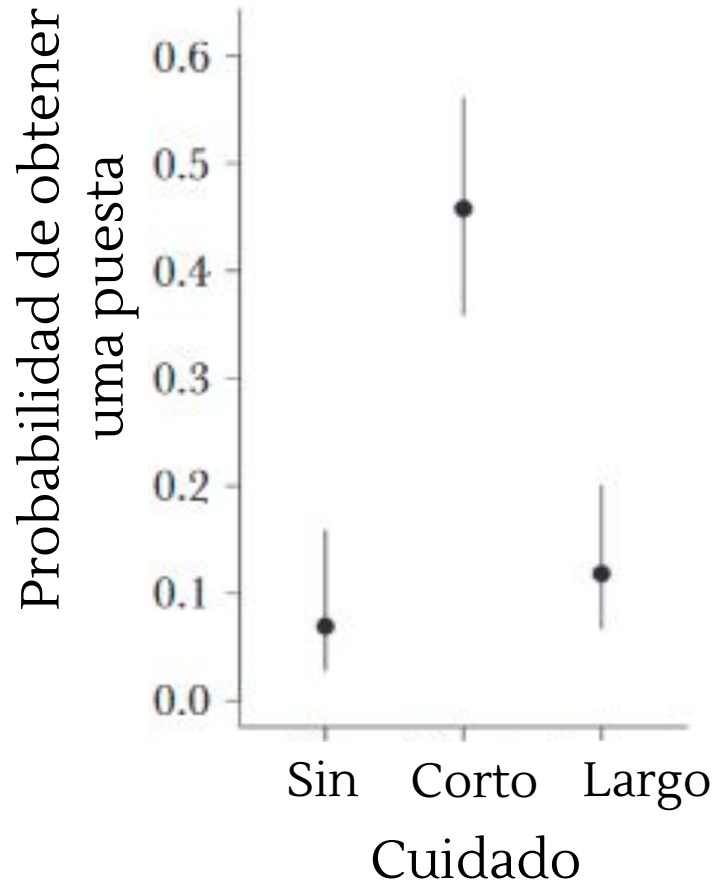
Behavioral Ecology (2015), 26(3), 926–935. doi:10.1093/beheco/arv035

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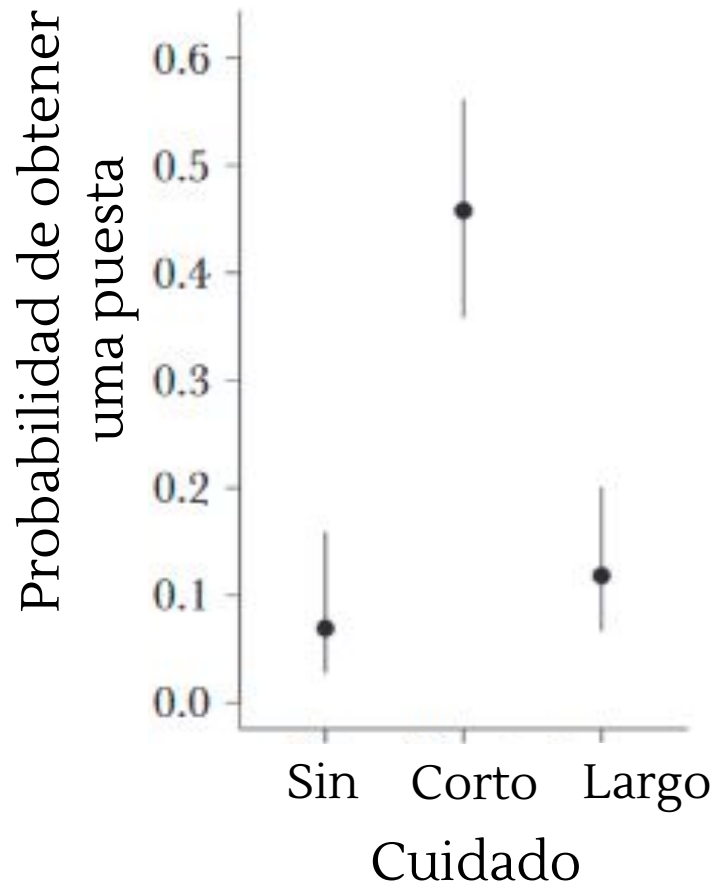


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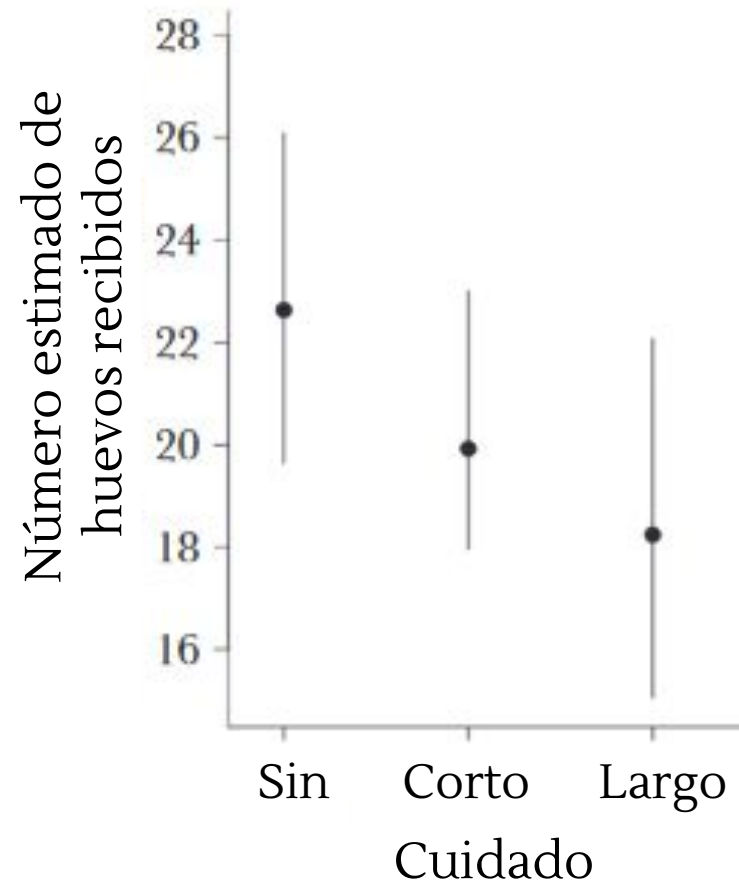
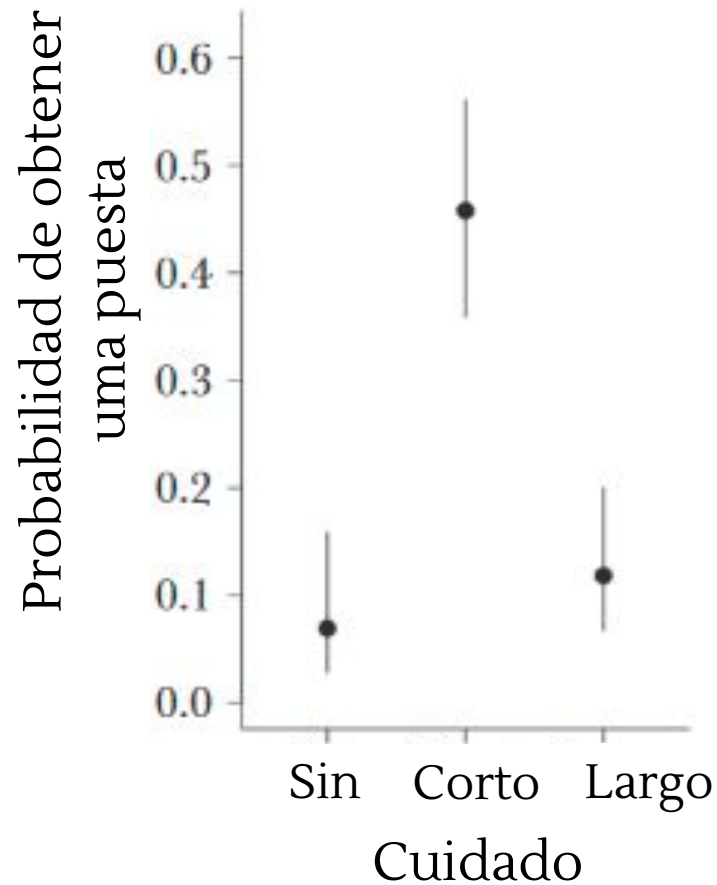


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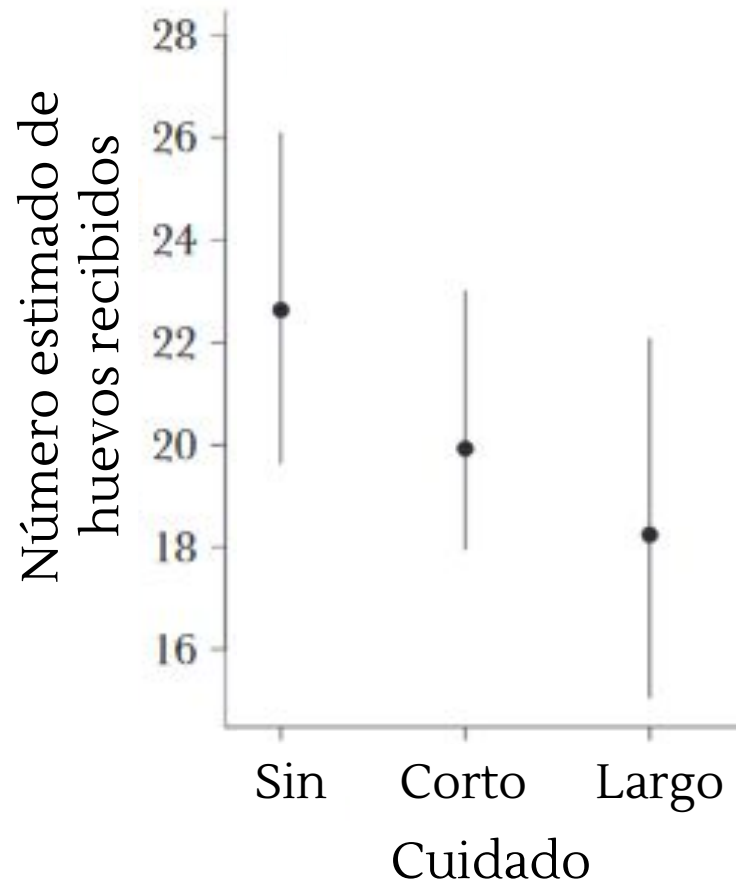
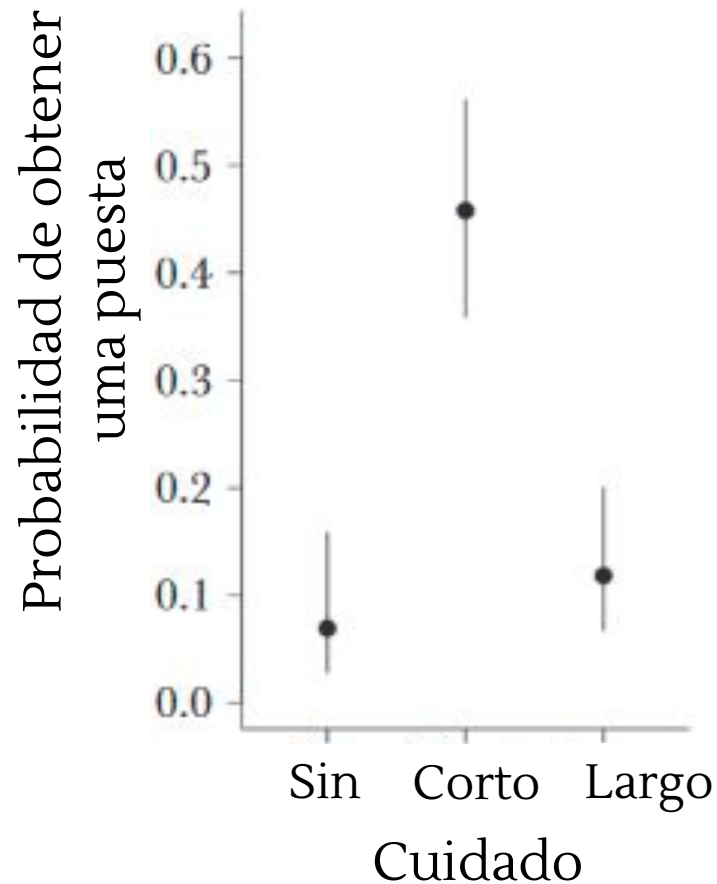


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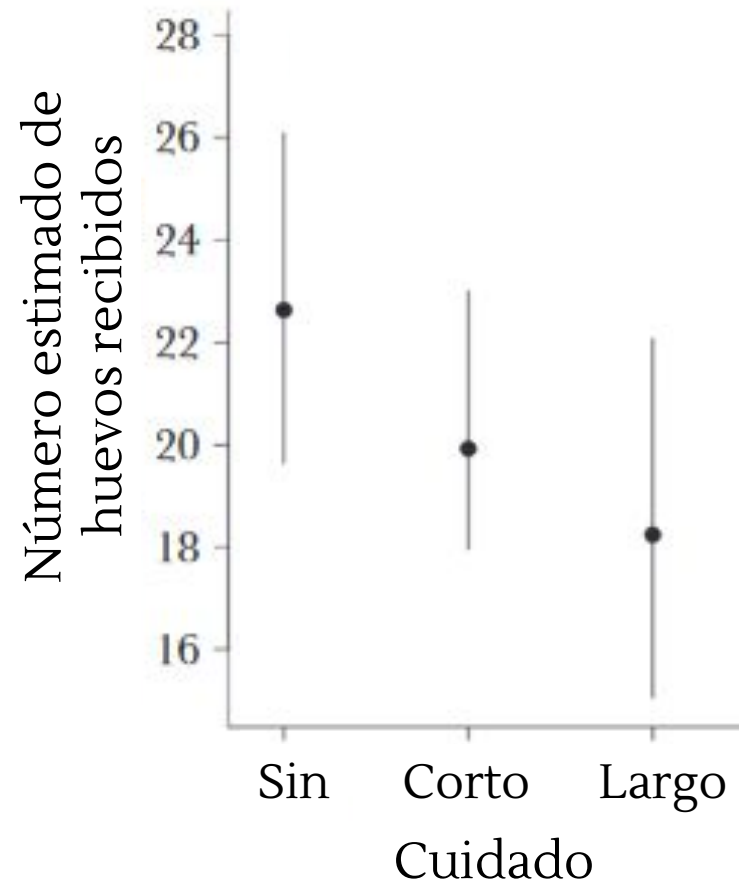
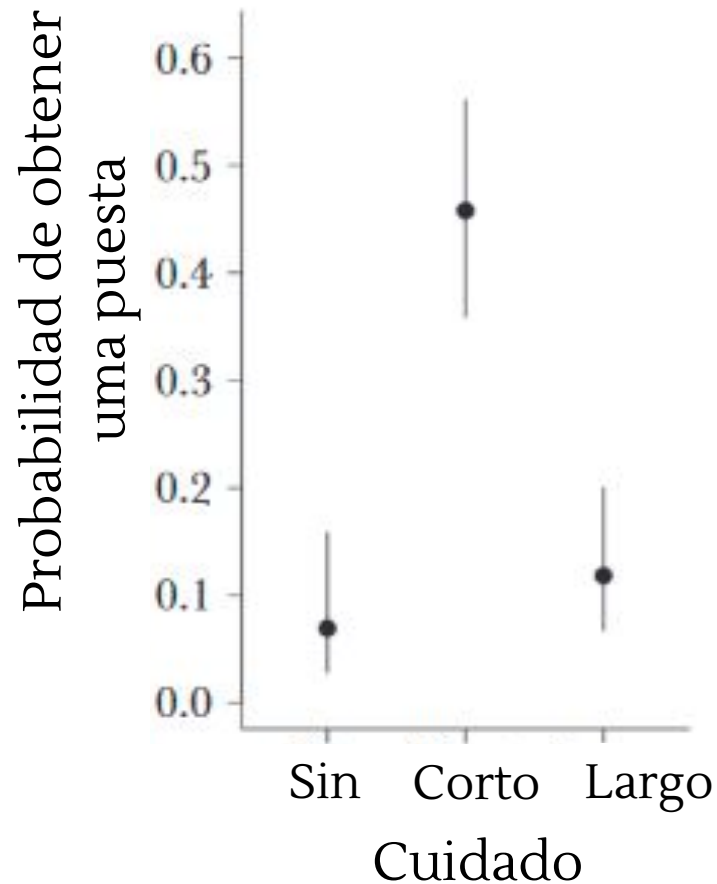
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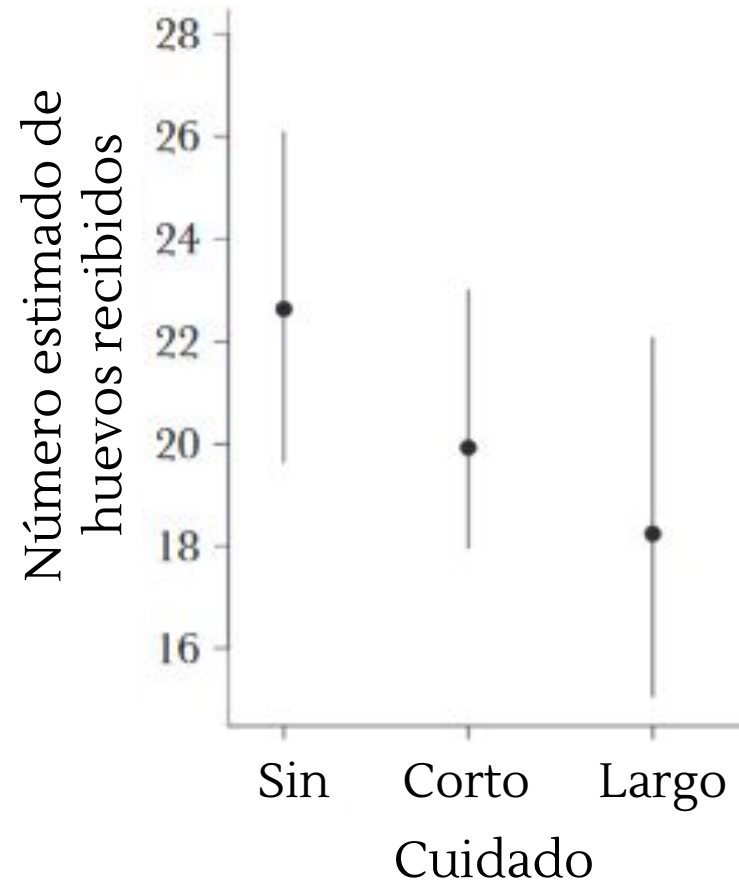
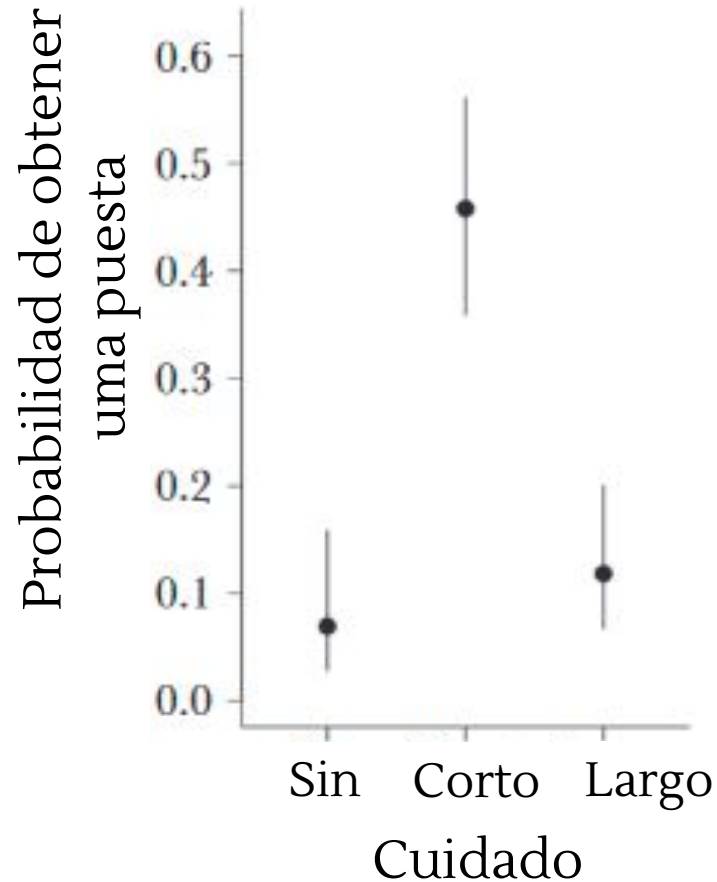


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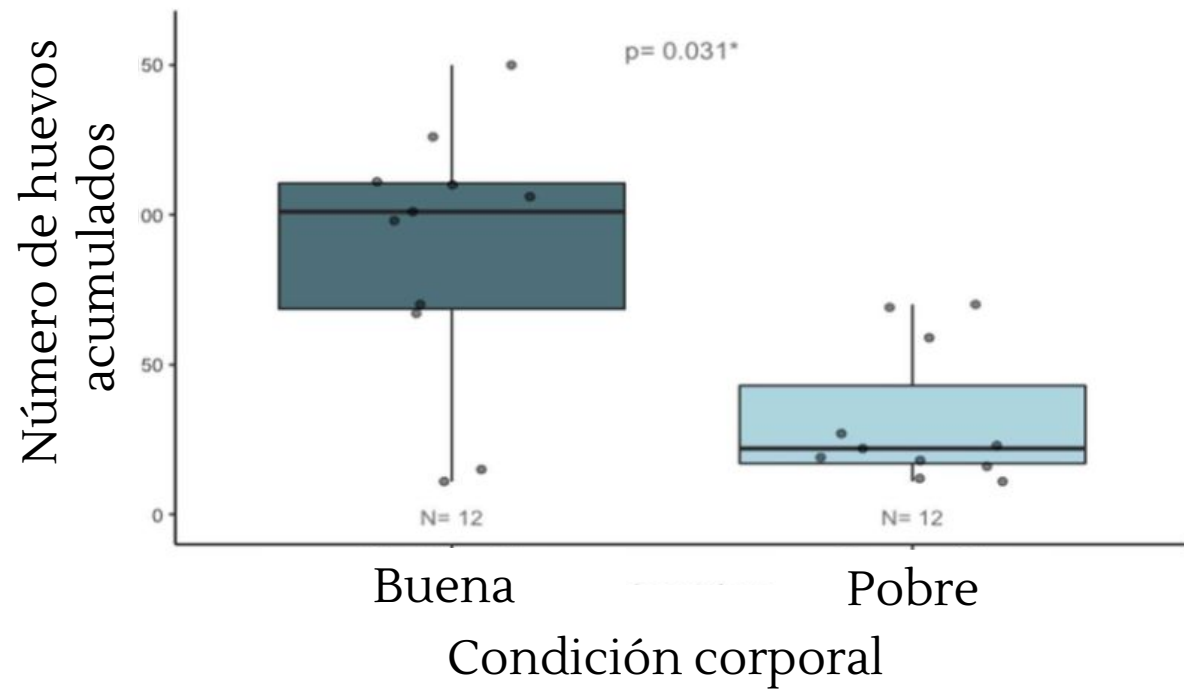


Good body condition increases male attractiveness but not caring quality in a neotropical arachnid with male-only care

Louise M. Alissa^{1,2}  · Glauco Machado³ · Gustavo S. Requena^{3,4}

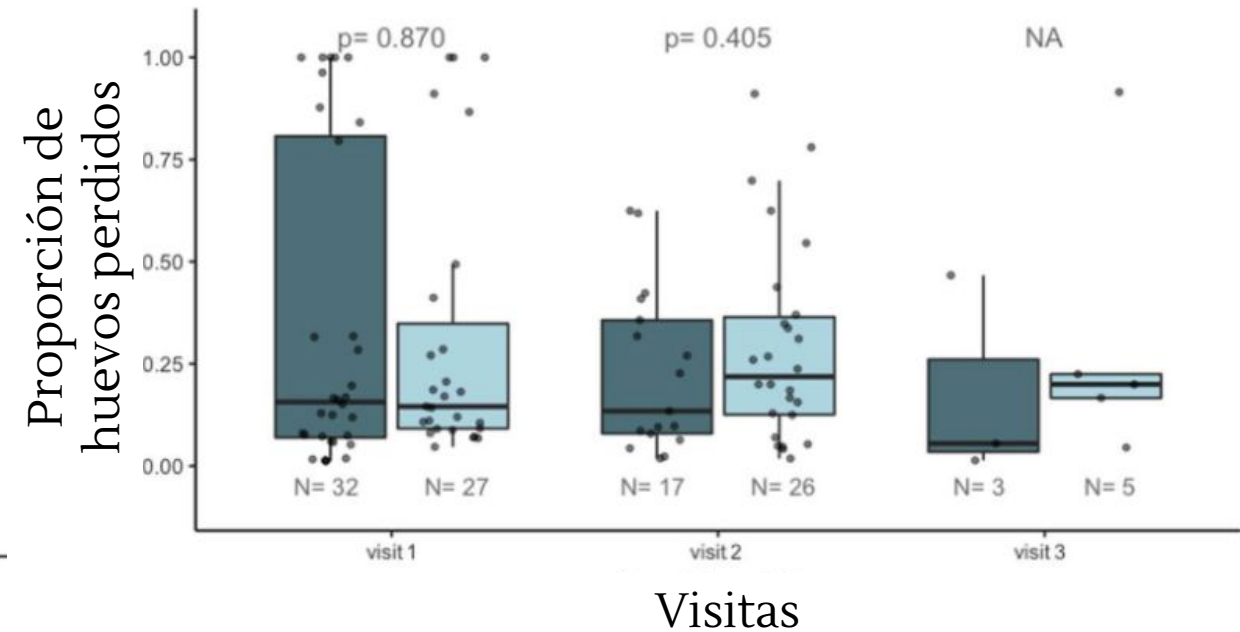
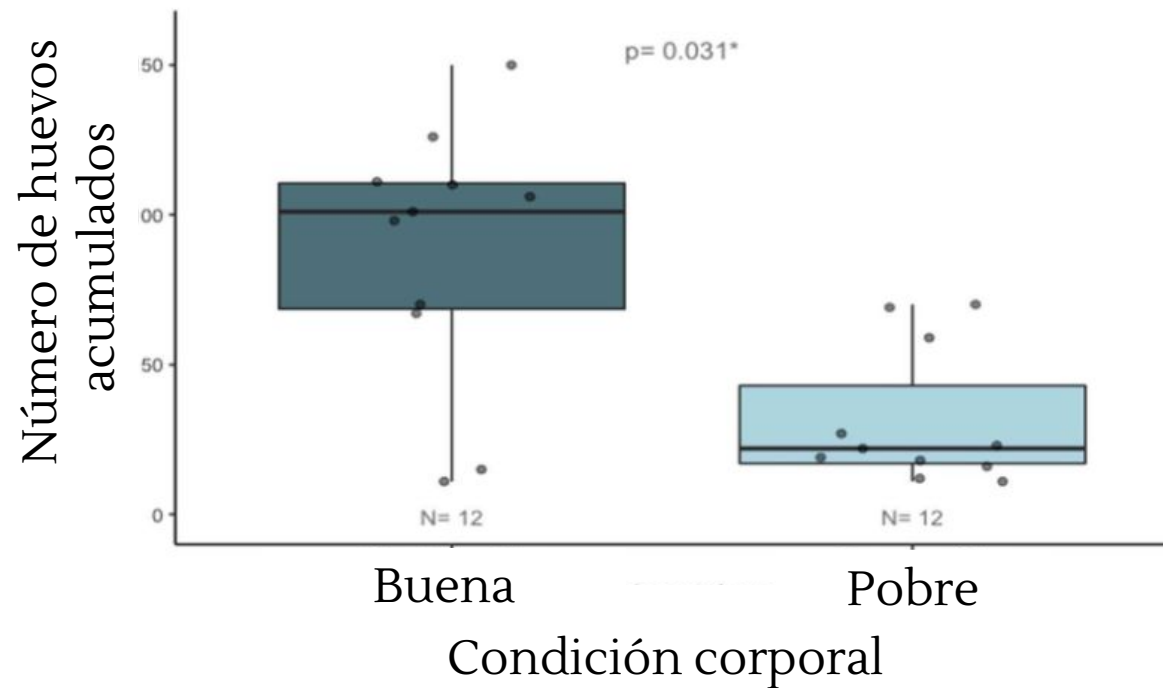
Good body condition increases male attractiveness but not caring quality in a neotropical arachnid with male-only care

Louise M. Alissa^{1,2}  · Glauco Machado³ · Gustavo S. Requena^{3,4}



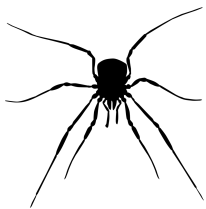
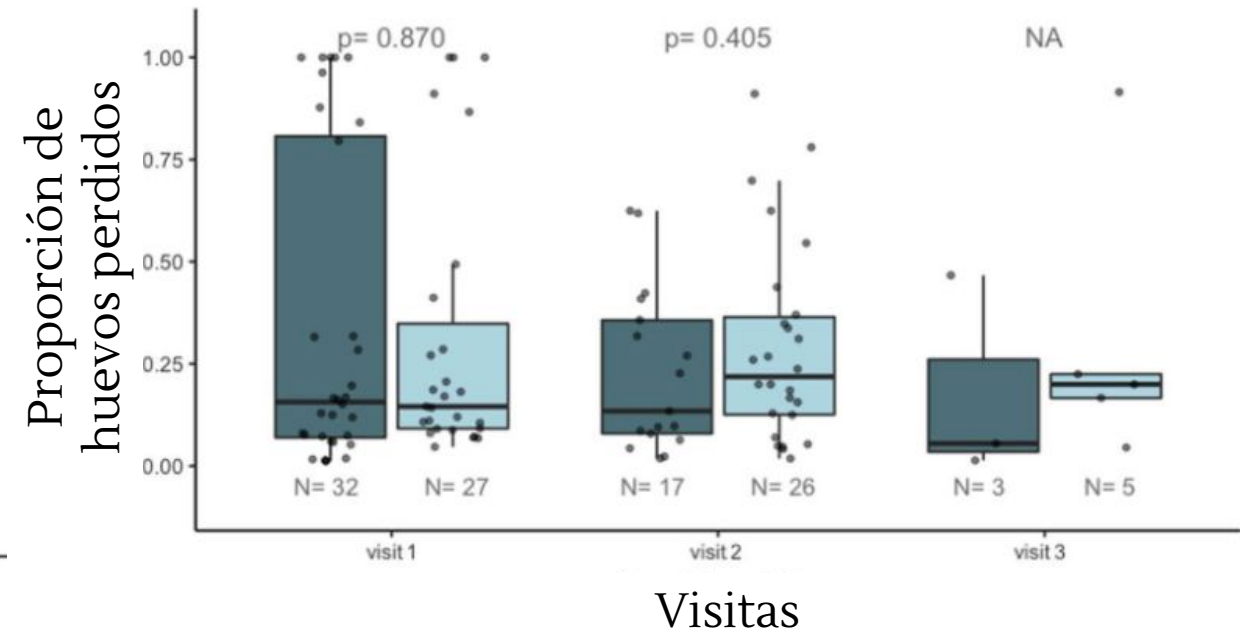
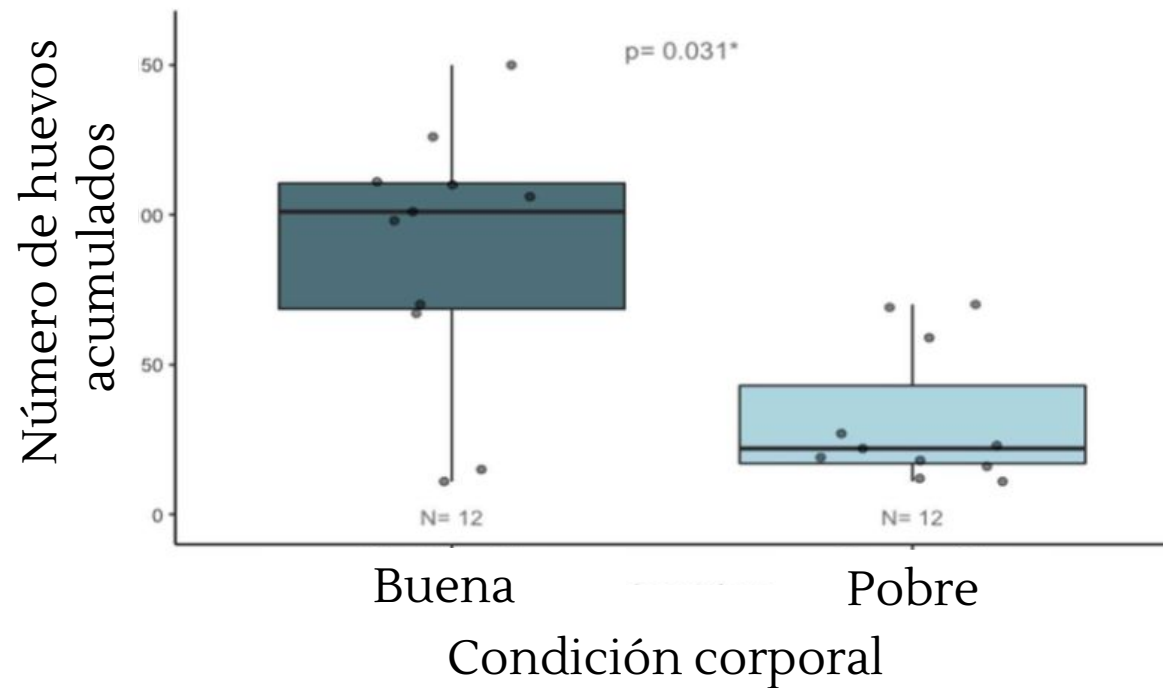
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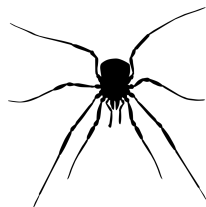
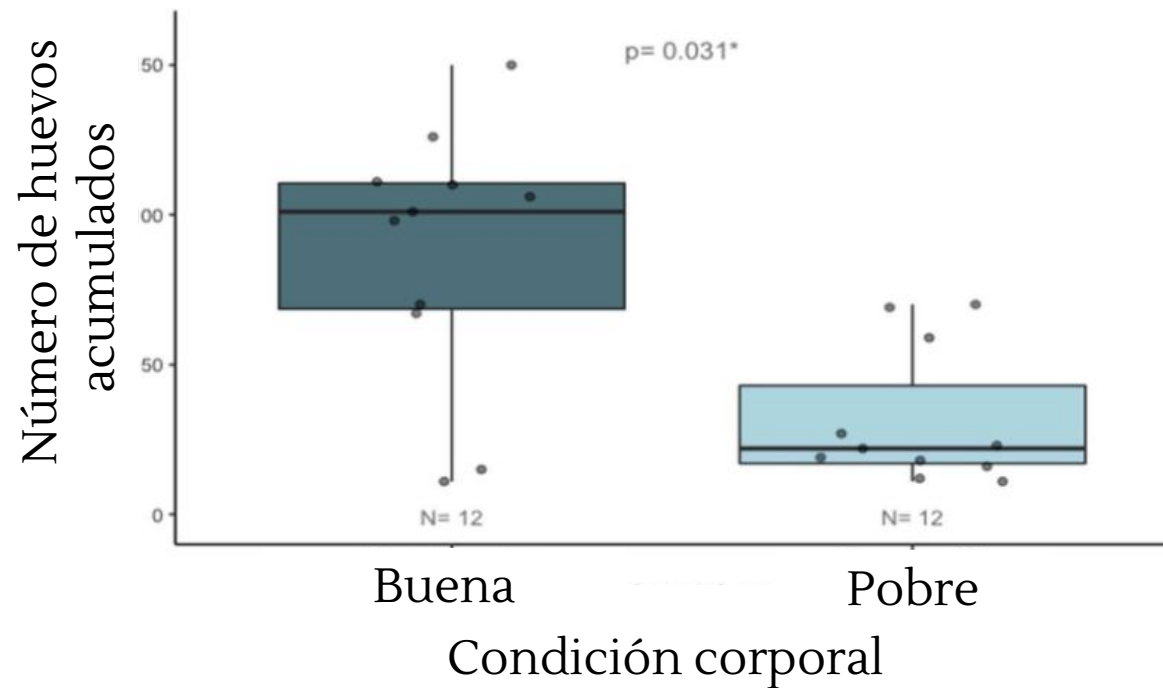
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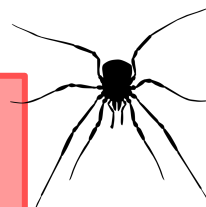
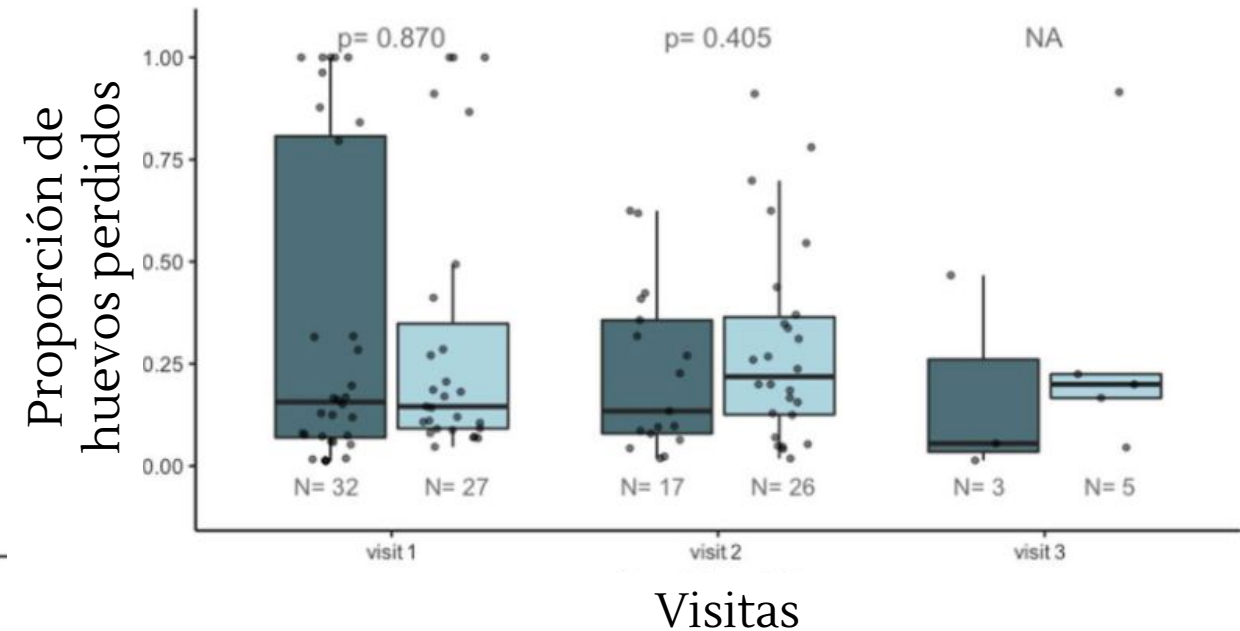
**Papás “en forma”
son sexys...**

Good body condition increases male attractiveness but not caring quality in a neotropical arachnid with male-only care

Louise M. Alissa^{1,2}  · Glauco Machado³ · Gustavo S. Requena^{3,4}



**Papás “en forma”
son sexys...**



**... pero no cuidan
mejor de los huevos**

Grossel & Machado,
submitted



Grossel & Machado,
submitted

Essential Male Care
(Kelly & Alonzo 2009)

Macho #1



Macho #2



Macho #3



CUIDADO

ANUNCIO

Grossel & Machado,
submitted

Essential Male Care
(Kelly & Alonzo 2009)

Macho #1



Macho #2

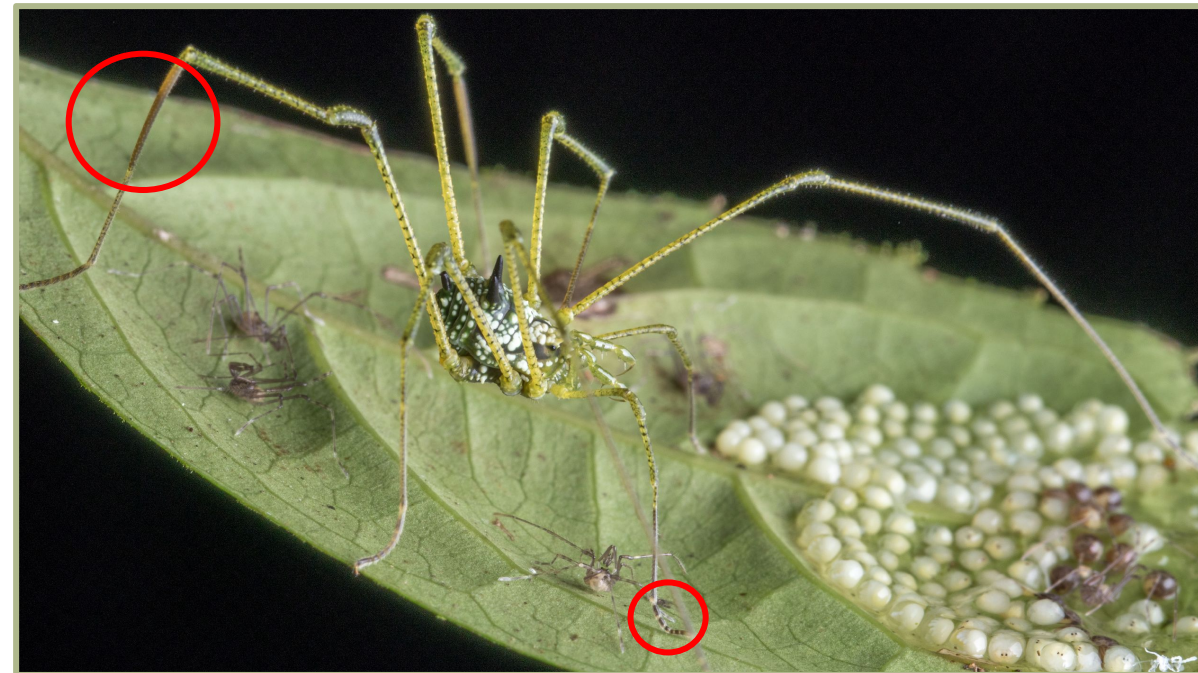


Macho #3



CUIDADO

ANUNCIO



Grossel & Machado,
submitted

Essential Male Care
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Macho #1



Macho #2



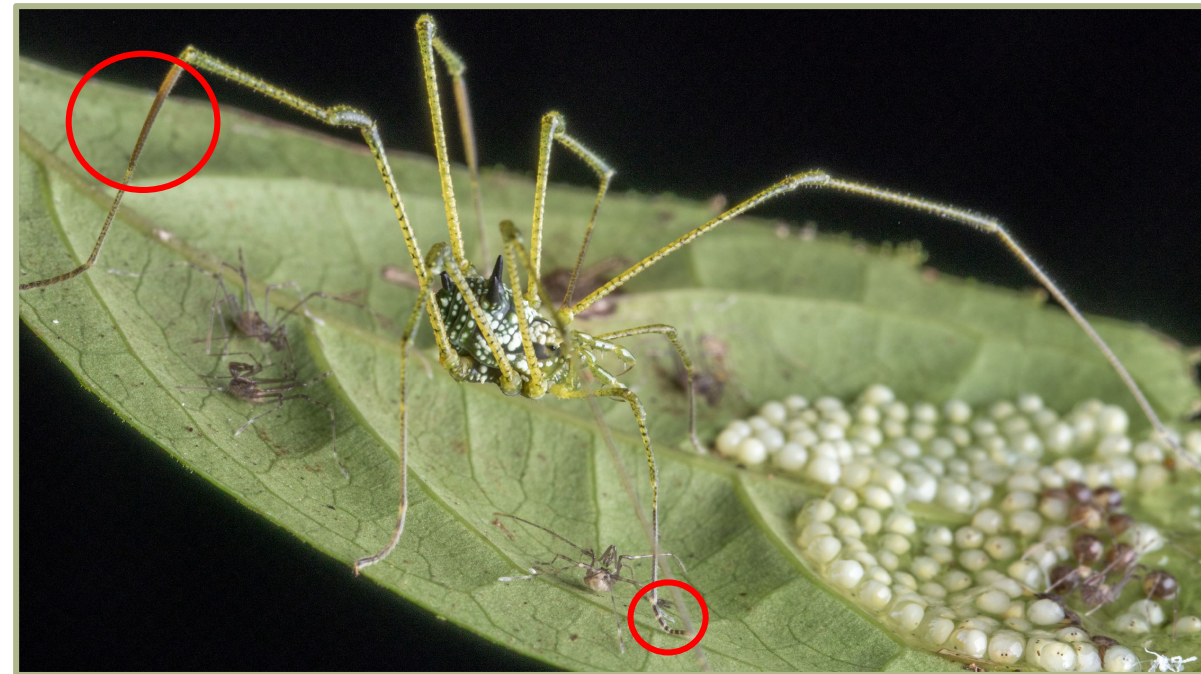
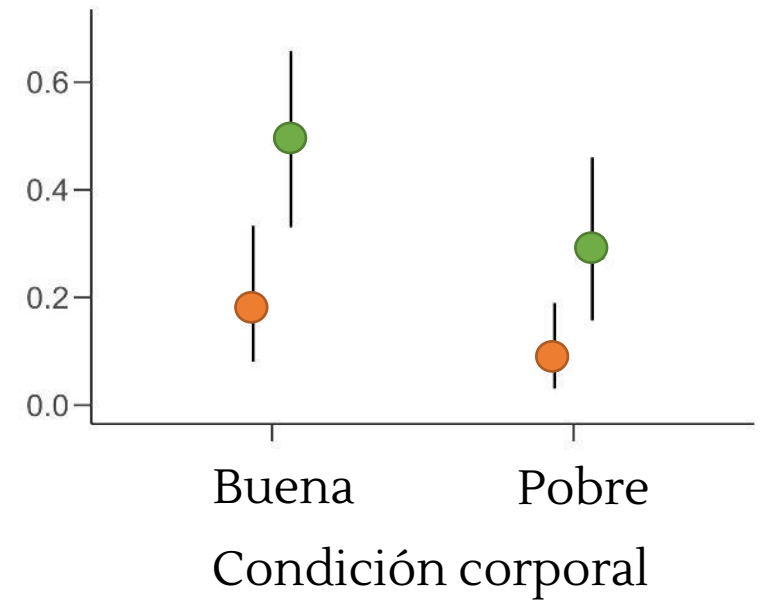
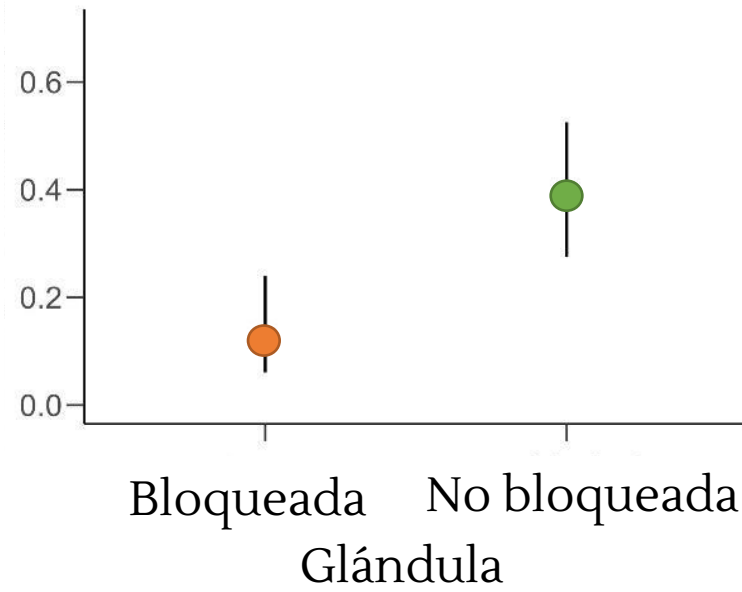
Macho #3



CUIDADO

ANUNCIO

Probabilidad de obtener
más huevos



Grossel & Machado,
submitted

Essential Male Care
(Kelly & Alonzo 2009)

Macho #1



Macho #2



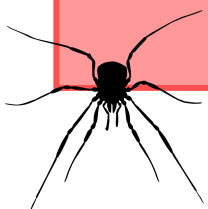
Macho #3



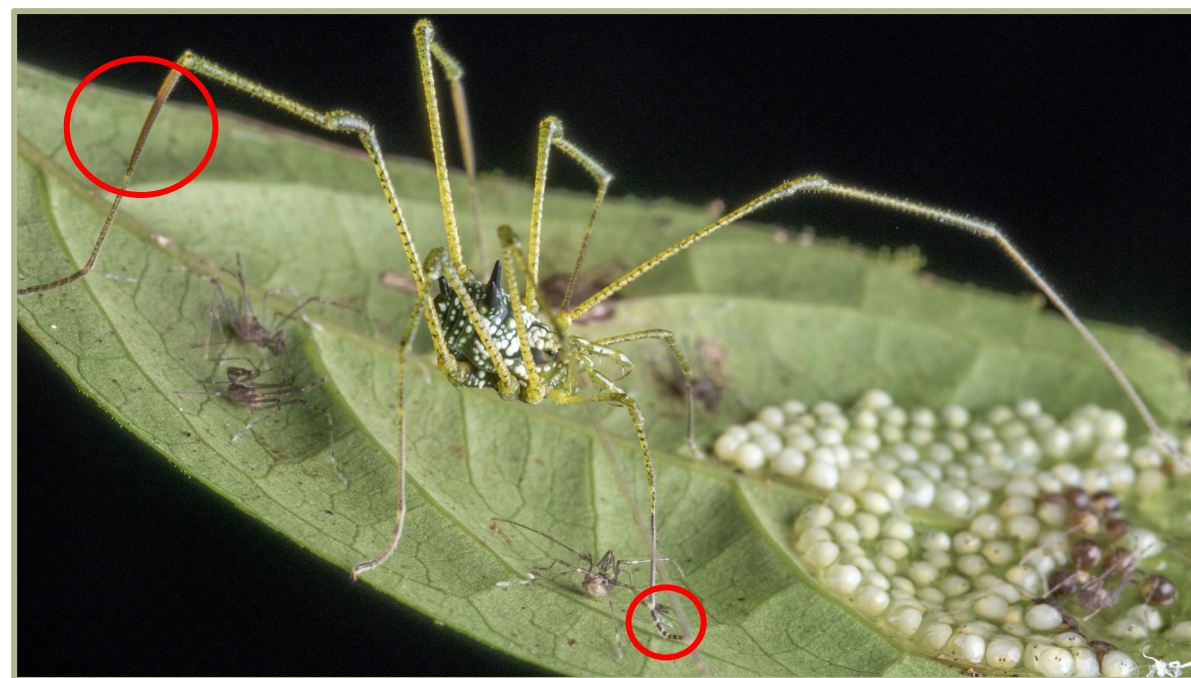
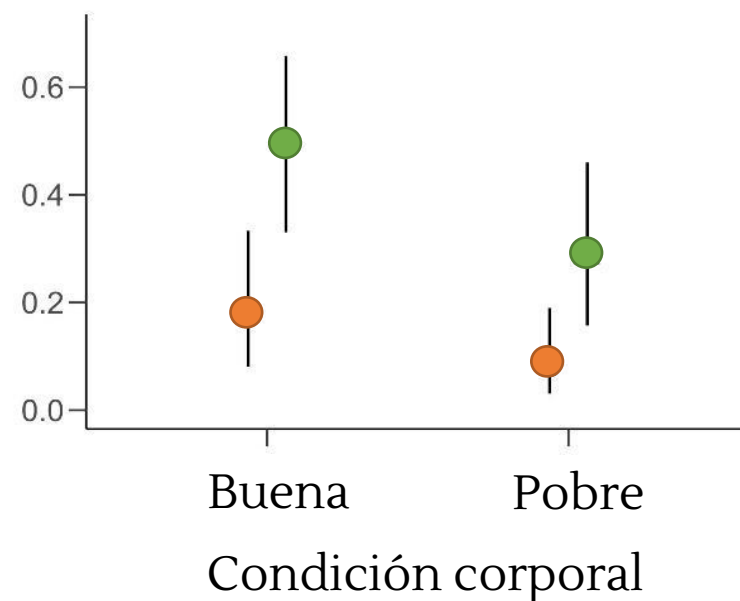
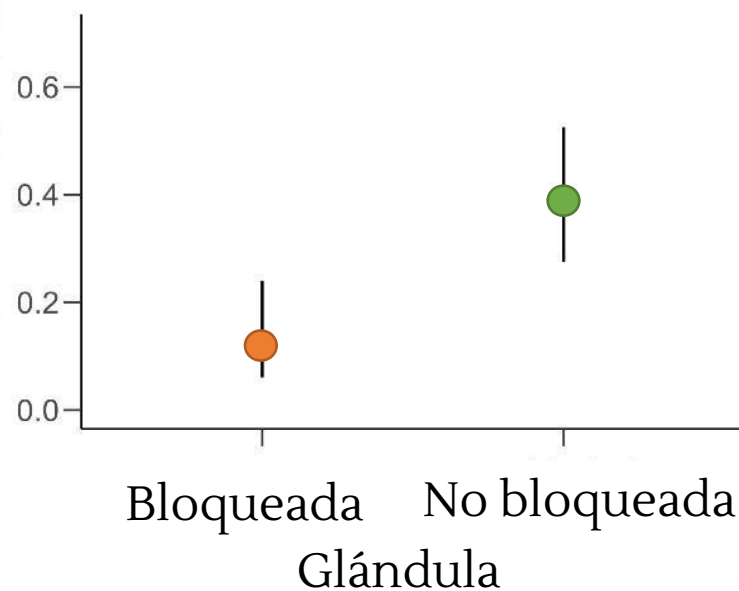
CUIDADO

ANUNCIO

Anuncio sexual es un indicador
de la calidad del macho



Probabilidad de obtener
más huevos



Lo que aprendimos de *Iporangaia pustulosa* durante 20 años:

2004

2009

2012

2014

2015

2024

2024?



Cuidado paterno
exclusivo



Lo que aprendimos de *Iporangaia pustulosa* durante 20 años:

El cuidado aumenta la supervivencia (**da puesta o del macho**)

2004

2009

2012

2014

2015

2024

2024?



Cuidado paterno exclusivo



Lo que aprendimos de *Iporangaia pustulosa* durante 20 años:

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Lo que aprendimos de *Iporangaia pustulosa* durante 20 años:

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Preferencias de las hembras por **machos cuidadores**, en **buena condición** y **que anuncian**

Lo que aprendimos de *Iporangaia pustulosa* durante 20 años:

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Cuidado paterno exclusivo



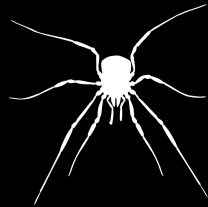
Preferencias de las hembras por **machos cuidadores**, en **buena condición** y **que anuncian**

¡Muchas gracias!

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Glauco Machado
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Liliam M. Rossetto
Louise M. Alissa
Roberto Munguía-Steyer

